Guaconomics: Avocado Price Shocks and Crime Rates in Mexico

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October 12, 2024

Abstract: Avocado trade between Mexico and the United States has exponentially burgeoned since 1997. Politicians have a growing concern about avocado production leading to more harm than good based on drug cartel involvement. Using data on 460 municipalities from 1997 to 2019, I exploit exogenous changes in U.S. weather deviations to show that increased Mexican avocado producer prices differentially decreased homicide in municipalities more climatically suited to grow avocados. This is because of increased agricultural employment from exponential growth in avocado production. However, there are different time heterogeneous effects. I find evidence that after the war on drugs took place in 2006, higher prices differentially increased homicide as well as other crime indicators such as property crime, cattle theft, and reported threats. The heterogeneous effects stem from increased drug cartel presence in the avocado industry.

Keywords: Development, International Trade, Agriculture

Acknowledgement: I would like to thank Joan Hamory for her invaluable feedback regarding the development of this paper. I thank Firat Demir, Tyler Ransom, Jayash Paudel, and Pallab Ghosh for their time, advice, and recommendations. I am also grateful to the University of Oklahoma's economics department and everyone who gave me feedback from seminars and brown bag presentations.

1 Introduction

The effects of international trade openness can vary depending on countries' idiosyncrasies. Trade openness can play an important role in creating jobs, increasing wages in both poor and rich countries, and improving working conditions (Newsfarmer, 2012; Brulhart, 2012). The Becker model shows that increased wages decrease the utility of committing a crime, making people less inclined to participate in illegal activity (Becker, 1968). These concepts combined indicate trade openness is likely to decrease crime, and related literature supports this notion (Prasad, 2012). However, little is known about the effects of trade openness and crime when drug cartels are involved. Powerful criminal organizations threaten to overwhelm local law enforcement institutions in a variety of regions, such as Latin America and Central Asia (Global Commission on Drug Policy, USDS 2012).

Mexico is the number one trading partner with the U.S. (Brookings, 2024). One of Mexico's most important agricultural products is avocados. According to the USDA 2019/2020 Annual Report, Mexico is the top avocado supplier to the U.S., with 87 percent of the market share in 2018. Herrera and Martinez-Alvarez (2022) shows that drug cartels actively exploit natural resources. Drug cartels diversify their economic portfolios into export-agricultural sectors such as avocado production. They diversify their portfolios because competition and state repression prompt them to look for non-traditional sources of income and to build up their violence-making capacities. Extracting rents from key industries allows drug cartels to establish territorial control in local communities. Unfortunately, drug cartel violence plagues Mexico. Through their Security, Justice and Peace department, Mexico announced in 2019, "The five most violent cities in the world are all Mexican: Tijuana, Juarez, Uruapan, Irapuato, and Ciudad Obregon". Figure 1 displays data from the World Bank that juxtaposes the intentional homicide rate per 100,000 people in Mexico and the world average from 2000 to 2019. In 2019, Mexico's homicide rate per 100,000 people was more than 300 percent larger than the world average. While policymakers become increasingly weary about the potential harms of drug cartel affiliation with avocado production, it is still unclear what the actual effect of the avocado market growth is on crime in Mexico. This is the first paper to provide a rigorous causal estimation of the impact of avocado price variation on crime in Mexico. This is also the first paper to exploit exogenous U.S. weather shocks that impact crime in Mexico through avocado prices.

Based on the methods of Dube et al. (2016), I evaluate the relationship between avocado prices and crime in Mexico. My empirical strategy exploits time variation in prices from weather shocks in California's avocado belt.[1] I interact cross-sectional variation in the agro-

[1] My empirical strategy closely follows Dube et al. (2016), who interact time variation of international

climatic avocado suitability of Mexican municipalities with avocado prices. My primary control for price is Mexico's real avocado producer price based on the methods of Nunn and Qian (2014). The most advanced system to assess agro-climatic suitability comes from FAO (FAO, 1983). However, this system requires extensive data for avocado suitability that does not currently exist for the studied region. In this paper, I develop a novel avocado municipality-level suitability index for avocado trees in Mexico's avocado belt by heavily applying methods from agricultural science (Dubrovina and Bautista, 2014; Gardiazabal, 2004; Lomas, 1988; Lovatt, 1990; Martínez, 1988; Rossiter, 1994; Tapia et al., 2007; Witten, 2005). The index accounts for both a municipality's soil and climate conditions. When interacting with Mexico's producer prices and the avocado suitability index via a difference-in-differences strategy, I determine whether avocado prices result in differential impacts on municipalities more agro-climatically suited for avocado production.

For this study, I create a panel dataset of 460 municipalities spanning 1997-2019 to assess the impact of avocado price changes on Mexican homicide. The sample includes municipalities in Michoacán and bordering states.[2] I study this region because Michoacán is Mexico's leading avocado-producing state and the only state with year-round production. From July 2018 to June 2019, Michoacán accounted for 76 percent of the national output.[3]. My estimates indicate that a 100 percent increase in Mexican avocado producer price decreases the impact of a unit increase of the suitability index on homicide by approximately 16 percent. Increased agricultural job opportunities from the growth in avocado production explain the negative relationship between producer price and homicide in Mexico.

My analysis also uncovers time heterogeneous effects between avocado price and Mexico's crime. In 2006, the war on drugs began in Mexico, and the government actively began attacking drug cartels. My analysis finds that from 2006 to 2019, an increase in avocado prices led to increased homicide. Additional data from Sistema Nacional de Seguridad Pública allows me to evaluate municipal-level crime in Mexico from 2011-2017. With this data, I find evidence that avocado producer price increases led to higher levels of property crime, cattle theft, reported threats, and homicide by gunfire. The related literature

corn prices and the FAO's maize suitability index in Mexico. They expose how external weather shocks from the U.S. corn belt impact corn prices and how the price change impacts illegal crop production in Mexico. This paper is related to studies that use cross-sectional variation in crop suitability, such as Nunn and Qian (2014), Qian (2008), and Qain and Nunn (2011), which focus on wheat, tea, and orchard cultivation, and potato suitability.

[2]Geographically, this consists of the following Mexican states: Michoacán, Jalisco, the state of Mexico, Colima, Guerrero, Guanajuato, and Querétaro.

[3]The USDA Avocado 2019/2020 Annual report further states that Mexico exported 186,958 metric tons of avocado to the U.S. Jalisco and the state of Mexico accounted for 9.2 percent and 4.5 percent, respectively. As of 2020, Michoacán was the only state in Mexico with phytosanitary clearance to export avocados to the United States.

supports the argument that these time heterogeneous effects heavily stem from cartels more aggressively diversifying their economic portfolios and gaining control of agricultural products such as avocados after the war on drugs began in 2006.[4]

I conduct several robustness checks to address potential threats to identification and measurement. One concern is that the effects from Michoacán may drive the results for two reasons. First, Michoacán produces the world's largest supply of avocado, and there is a lot of territorial contestation among drug cartels in the state. Second, since the implementation of NAFTA in 1994, there have been multiple occasions of increased avocado trade openness between the U.S. and Mexico.[5] Michoacán was the only state allowed to export avocados during the study. Policies that open trade can impact the price of avocados. To address whether trade openness is the only main cause of price fluctuation as opposed to California weather shocks, I omit Michoacán from the analysis, and the results are robust.

My findings remain robust to multiple weather deviation instrument combinations and account for nonlinear weather deviation. I also test my empirical findings using an alternative IV method from Nunn and Qian (2014) that provides consistent results. I further account for multiple price indicators and El Niño to ensure the results' validity. To test the validity of the methods to construct the suitability index, I apply two alternative constructions of the index, and the results remain robust.

Few papers have focused on how avocado trade with the U.S. and Mexico has impacted crime. That is because there is currently no baseline method to test how avocado production has a causal relationship with crime in Latin America. Erickson and Owen (2020) show through a one-time trade shock that avocado production quantity increased and led to lower homicide. On the other hand, Herrera and Martinez-Alvarez (2022) focuses on a broader group of Mexican agricultural export commodities and shows that access to primary sector revenues is associated with higher levels of violence among Mexican municipalities. My paper deviates from previous analysis by using a rigorous causal empirical method to show how avocado price variation impacts Mexican crime at the municipality level over different periods. This paper also deviates from this literature by using agricultural science to develop a novel suitability index. A standard municipal-level avocado index does not currently exist. The index allows me to analyze an exogenous measure of how suitable a

[4]Castillo (2014) points out that drug trafficking is no longer the main source of income for some criminal organizations in Mexico. Many drug cartels in Mexico make more from legal goods than they do illicit products such as marijuana and opioids.

[5]There have been numerous trade openness updates between the U.S. and Mexico after NAFTA's implementation in 1994. In 1997, the U.S. began actively importing avocados from Mexico to 19 Northeastern and Midwestern states and Washington D.C. In 2003, all but California, Florida, and Hawaii allowed avocado imports from Mexico. The U.S. allowed Mexican avocado imports to all states in 2007 (Carman, 2019).

municipality is to grow avocados.

This paper contributes to the literature on income shocks and conflicts.[6] Papers have focused on income inequality and labor wages (Briceño-Léon, 2008). Many studies find that higher income may increase conflict by promoting predation over resources (Hirshleifer 1991; Grossman 1999; Fearon 2005; Mitra and Ray 2014; Nunn and Qian 2011). Other research has found that higher income may decrease crime due to the increased opportunity cost of fighting (Becker 1968; Grossman 1991; Collier and Hoeffler 1998; Fearon and Laitin 2003; Miguel et al. 2004; Besley and Persson 2011; Do and Iyer 2010, Hidalgo et al. 2010, Gwande et al. 2012).

Similar to the research on income shocks and conflict, research on commodity prices and conflict demonstrates both negative and/or positive effects depending on the commodity and country. The relative strength of the opportunity cost and predation channels of a commodity may stem from labor intensity (Dal Bó and Dal Bó, 2011). Numerous studies show a negative relationship between conflict and export price (Brückner and Ciccone 2010; Berman and Couttenier 2013; Bazzi and Blattman 2014). Another side of the literature shows a positive relationship between conflict and price (Maystadt et al., 2014; Besley and Persson, 2008 and 2009). However, work such as Dube and Vargas (2013) shows both effects can take place in the same empirical setting. My analysis differs from this literature by closely following the methods of Dube et al. (2016), but I evaluate avocado instead of corn. The heterogeneous period effects match many of the varying effects the literature discusses. To my knowledge, this paper is the first to show that U.S. weather shocks in California's avocado-producing region impact crime in Mexico. This analysis clarifies the complicated nature of avocado production in Mexico and the varying effects of trade with the U.S. My findings warrant caution about the disturbing development of increased avocado production in Mexico, a nation torn by violence.

The remainder of this paper is organized as follows: section 2 includes a background on drug cartel violence in Mexico and avocado production and trade between the U.S. and Mexico; section 3 describes the analysis's data; section 4 explains my empirical strategy; section 5 discusses my results; and section 7 concludes.

[6]Violence in Mexico comes from many sources among international trade. For example, Dell et al. (2019) shows that loss of manufacturing jobs from competition with China increased cocaine trafficking and violence in Mexico. Criminal groups exercise risk-reduction and alternative investment risk strategies when the government threatens their traditional revenue sources.

2 Background

This section provides a background on two important categories of this paper. First, I discuss the evolution of Mexico's cartel violence. Second, I describe the development of the avocado industry in Mexico and the U.S. over the past 30 years

2.1 Mexican Drug Cartels

The rising U.S. demand for marijuana in the 1960s dramatically increased the Mexican drug trade. Both Mexico and Colombia worked together to traffic cocaine from South America (Astorga, 2005; Toro, 1995). In the 1980s, underlying political conditions in Mexico led to restrained violence. In the absence of political competition, the PRI political party consolidated the patron-client relationship between drug traffickers, the police, and local elected officials, enabling cartels to operate in particular locations with relative impunity (O'Neil, 2009). The entry of political parties in local elections in the 1990s undermined these arrangements and led to territorial expansion and inter-cartel fights (Bartra, 2012; O'Neil, 2009; Osorio, 2012). While Mexico's crime increased in the 1990s, it skyrocketed in the 2000s. In December 2006, President Felipe Calderón launched an aggressive military campaign against the drug cartels known as Mexico's war on drugs. This political action resulted in dramatic and haphazard spikes in violence throughout the nation (Berman, 2015). The drug war has inflicted rising violence in both urban areas and rural areas with engaged drug crop cultivation (Escalante, 2009). Fighting among cartels in Michoacán has been linked to attempts to control production areas and routes (Maldonado Aranda, 2012).

Beittel (2015) states the four major drug trafficking organizations (DTOs) as of 2006 were the following: the Tijuana/Arellano Felix organization (AFO), the Sinaloa cartel, the Juarez/Vicente Carillo Fuentes organization (CFO), and the Gulf cartel. The Tijuana/Arellano Felix organization is based in the border city of Tijuana and the Sinaloa cartel is in Sinaloa. The Juarez/Vicente Carillo Fuentes organization is based in the border city of Ciudad Juarez, which is in Chihuahua. Lastly, the Gulf cartel is based in the border city of Matamoros, Tamaulipas. Since 2006, the U.S. DEA has identified additional major cartels such as Beltran Leyva and La Familia Michoacána. Beltran Leyva and La Familia Michoacána are in Sonora (a border state) and the more centrally located state of Michoacán, respectively. Estimates guess these organizations may have fragmented into as many as 20 major organizations over the last several years. With the growth of more major criminal organizations, more brutal homicide rates continued to increase. The Mexican people publicly become aware of larger volumes of homicide through beheadings, public hanging of corpses, killing innocent bystanders, torture, devastating car bombs, and even assassinations of journalists and government officials.

2.2 The Evolution of Avocado Prices

The increase in avocado prices can be heavily attributed to the change in public perception of avocados as healthy. Carman (2019) provides a rich history of U.S. demand for avocados. The U.S. per capita consumption of avocados averaged 1.51 pounds in the 1990s. During the 1990s, California dominated the U.S. avocado market; fresh imports typically accounted for less than 1 percent of total U.S. consumption. U.S. consumption per capita dramatically skyrocketed to 8 pounds in 2018. The change in U.S. demand stems from successful marketing and public perception of avocados. The CAC began to fund diet and nutritional research to proactively inform the public of the health benefits of eating avocados. This research led the U.S. public to perceive avocados as a component of a healthy diet.

Progressive trade openness has also impacted avocado prices. In 1994, the North American Free Trade Agreement (NAFTA) was enacted. In February 2007, the U.S. opened avocado trade with Mexico in all 50 states. Initially, the USDA only permitted avocado imports from the state of Michoacán. In 2016 the USDA amended the Mexican Hass Avocado Import Program to allow all states that met phytosanitary guidelines to export their avocados to the U.S.

Figure 2 shows both the logged real producer price and logged non-FAS converted real import price of avocado from Mexico to the U.S. Data for nominal producer price and import price come from the Food and Agriculture Organization of the United Nations (FAO, 2024) and the USDA's Global Agricultural Trade System (GATS, 2024).[7]. Both the import price and Mexican producer price follow similar trends. There is a large spike in the Mexican producer price from 1996-1999, likely due to NAFTA's enactment in 1994 and the U.S. opening avocado trade in 1997. From 2003 to 2007 there is a gradual increase in both price measures. This is likely due to further trade openness between the U.S. and Mexico in these years. The price increase in 2007 was likely caused by California experiencing an unexpected January frost that destroyed nearly 3 quarters of the state's avocado supply that year (NPR, 2007). The visible increase in import prices from 2015 to 2019 is expected. California experienced heat waves in 2015, 2016, and 2018 that damaged California's avocado supply (Gambino, 2015; Ferguson, 2016; California Avocado

[7]Based on the work of Nunn and Qian (2014), I convert all nominal prices to real prices by using U.S. CPI and making 1996 the base year (Federal Reserve, 2024)

Commission, 2018). This price increase is also likely due to the amendment of the 2016 Mexican Hass Avocado Import Program.

3 Data

This section discusses 3 different data sets this paper utilizes and the construction of my novel avocado suitability index. INEGI's municipal mortality data stretches back to 1990. Because of the richness of this data, my main analysis focuses on the impact of avocado price variation on homicide from 1997 to 2019. In addition, I use census data from INEGI's 1990 and 2000 Census and Countings to evaluate how increased avocado production and trade impacted Mexico's agricultural jobs. Lastly, this paper explores time heterogeneous effects that avocado prices have on crime in Mexico. Mexico's Secretariado Ejecutivo del Sistema Nacional de Seguridad Pública (SESNSP) contains municipal-level crime data for a larger number of crime categories from 2011 to 2017. I utilize SESNSP's data to explore the issue further when evaluating the time heterogeneous effects of avocado prices.

3.1 Main Analysis Data

Data for municipal-level homicide come from INEGI's Registered Death Statistics. The data includes every recorded death's time, location, and cause in Mexico from 1990 to the present.[8] To measure homicide, I add all deaths from INEGI that resulted from homicide at an annual and municipal level from 1997 to 2019. I chose this sample because Mexico began exporting avocados to the U.S. in 1997, and data after 2019 would be heavily biased from the COVID-19 pandemic. To ensure that my analysis only compares rural municipalities, I drop all municipalities with a population above 100,000 from the sample based on the methods of Dube et al. (2016).

INEGI's General Census of Population and Housing for the years 1990, 2000, and 2010 has data on each municipality's population, the percent of the population that has completed primary education, the male population, and the number of economically active persons.[9]. I match the census data with the following decades' homicide values at the municipal level.[10]

[8]INEGI's death records are publicly available on their website and can be found at the following website: https://en.www.inegi.org.mx/programas/edr/

[9]INEGI Conducts their General Census of Population and Housing every ten years. Their census micro-level data is publicly available on their website: https://en.www.inegi.org.mx/programas/ccpv/2000/

[10]For example, I match the 1990 census data with homicide observations at the municipal level for the years 1997-1999, the 2000 census data for the years 2000-2009, and the 2010 census data for the years 2010-2019.

I base my weather data collection on the methodology that Dell et al. (2014) discusses. Cross-sectional data for the 1980-2010 municipal temperature in Celsius and precipitation in inches averages come from Mexico's Servicio Meteorológico Nacional (SMN). [11]. SMN's data comprises of municipalities' weather station averages from 1980-2010. For my analysis, I take the annual average temperature and precipitation of all municipal weather stations for these 30 years. For municipalities with no weather station data, I take the average of weather stations within all contiguous municipalities.

Annual-level variables include real Mexican producer price (USD/Ton), real FAS converted (USD/KG) and real non-FAS converted import price (USD/Ton), average California temperature deviations (degrees), average California precipitation deviations (inches), and average minimum January temperature deviations (degrees).[12] Based on the methods of Nunn and Qian (2014), I convert all nominal price data into real price by using U.S. CPI, where 1996 is my base year. Producer price comes from the FAO, while import prices are from GATS.[13].

California's average weather data comes from the National Centers for Environmental Information (NCEI). Avocados in California are most at risk from overheating in April-July. In addition, avocado trees are the most vulnerable to precipitation in April-July. For both average temperature and precipitation, I average all avocado-producing counties data from April-July. Avocados are the most at risk of frost in January. To measure the minimum average January temperature, I take the average minimum temperature of that month among all California avocado-producing counties each year. To calculate weather deviations, I take the difference between this region's 1900-2000 weather average and the annual weather average. Table 1 shows the summary statistics for my main analysis. When I combine all of these variables, the panel data includes 460 municipalities spanning 1997-2019 and 10,285 observations. [14]

3.2 Index

The standard agriculture suitability indexes come from FAO. FAO does not have an avocado suitability index at the municipal level. In general, there is no well-recognized avocado

[11]Data for all weather stations come from CONAGUA and are publicly available through Gobierno de Mexico: https://smn.conagua.gob.mx/es/informacion-climatologica-por-estado?estado.

[12]Producer price represents the amount of money paid to Mexican farmers in USD/Ton. FAS-converted import prices account for transportation costs, insurance during transit, and import duties.

[13]Mexico's nominal avocado producer prices are publicly available on FAO's website: https://www.fao.org/faostat/en/. In addition, annual avocado import prices from Mexico to the U.S. are publicly available on the following website: https://apps.fas.usda.gov/gats/ExpressQuery1.aspx

[14]County-level weather data are publicly accessible on the following website: https://www.ncei.noaa.gov/access/monitoring/climate-at-a-glance/county/mapping.

suitability index that exists. In this paper, I create a novel avocado suitability index for all municipalities in Michoacán and states bordering Michoacán. I use agricultural science methodology by closely following the work of Dubrovina and Bautista (2014) to develop the index. The index focuses on two components of a municipality's suitability for avocado trees: soil and climate.

Avocado tree roots are susceptible to soil types. They require both a large volume of water and low hydraulic conductivity. To compute the suitability of a municipality's soil, I take advantage of INEGI's detailed edaphology data to find out the predominant soil in each municipality.[15]. Applying soil characteristics discussed in Dubrovina and Bautista (2014) and ISRIC World Soil Information, I can classify whether the soil is one of the following: Highly suitable, Suitable, Low Suitability, and No Suitability.

Figure 3 displays characteristics that constitute a soil's avocado suitability. A combination of a base saturation percentage below 42.4 percent and a clay content below 20 percent results in the most suitable soil type for avocado trees.[16] Other important factors impacting a soil's avocado tree suitability are its depth, aggregate shape, hydraulic conductivity, and clay content. Avocado trees grow best in soil with a depth greater than 48 cm. Soils that are angular blocky, granular, and/or powdery are suitable, while massive aggregate shapes result in no suitability. If the soil has a high base saturation, deep depth, and a crumb aggregate shape, then the soil's hydraulic conductivity and/or clay content decides its avocado suitability. In this case, low hydraulic conductivity is suitable for avocado trees, but very high hydraulic conductivity results in low suitability. In addition, if the hydraulic conductivity is high and the clay content is equal to or less than 28 percent, the soil offers low suitability. However, if the clay content is above 28 percent, then the soil is suitable. If the municipality's most common soil is highly suitable, then I give it a soil value of 3, I give the municipality's soil a 2 value if its suitable, 1 if the soil has low suitability, and 0 if the soil has no suitability for avocado trees. I denote soil S in municipality i as S_i. Table A3 in the appendix provides the general characteristics and index values of dominant municipality soils in my sample.

The second portion of the index takes into account a municipality's climate. Key climate components of a municipality's avocado suitability include mean annual temperature (Celsius), the ratio of precipitation (inches) to mean annual temperature (P/T), altitude

[15]INEGI's edaphology data is publicly available on the following website: https://en.www.inegi.org.mx/temas/edafologia/

[16]Avocado trees require very distinctive conditions that are uncommon worldwide. Andosol is the only highly suitable soil based on the methods of Dubrovina and Bautista (2014). It is also the world's rarest soil. Michoacán has a large landmass of Andosol soil that helps make it arguably the most conducive climate for avocados worldwide.

(meters), and average temperature in January. Weather averages for mean annual temperature, precipitation, and the month of January at the municipal level come from SMN's 1980-2010 weather station averages. A municipality's altitude range comes from INEGI's edaphology data.

Figure 4 shows how I determine a municipality's avocado climate suitability value. Like my soil calculations, a municipality can have one of the following values for avocado suitability: Highly Suitable, Suitable, Low Suitability, and No Suitability. Avocados are highly sensitive to frosts, heat waves, and water volume. A mean annual temperature equal to or below 11.9 Celsius is not suitable for avocado trees. If a municipality's mean annual temperature is above 11.9 Celcius and its P/T is below 40.6, then it has low suitability. In the same case, if its P/T value is above 40.6 and its altitude is above 2,505 meters, it has low suitability.[17] A high mean annual temperature, high P/T value, altitude below 2,505 meters, and an average January temperature equal to or below 15.7 would lead to a suitable climate. Otherwise, if a municipality's average January temperature is above 15.7 Celcius, the climate value depends on the magnitude of the P/T ratio. A P/T value, in this case, above 53.9 would result in a highly suitable climate, while a value below or equal to 53.9 represents a suitable climate. I give a value of 3 to a municipality with a highly suitable climate, 2 to a suitable climate, 1 for low suitability, and 0 for no suitability. I denote climate C in municipality i as C_i. Equation 1 displays the conditional function of a municipality's avocado suitability index in municipality i.

$$Index_i = \begin{cases} \mathrm{S}_i + C_i & \text{if } \mathrm{S}_i \neq 0 \text{ and } \mathrm{C}_i \neq 0 \\ 0 & \text{if } \mathrm{S}_i = 0 \text{ or } C_i = 0 \end{cases} \tag{1}$$

The values for S_i and C_i range from 0 to 3. If both S_i and C_i do not equal 0, then the avocado suitability index value for a municipality will be their sum. If either S_i or C_i have a value of 0 then the index value is also 0. This is the case because even if a municipality has the best climate suitability, if its soil is not suitable for avocado growth then it is still unsuitable for avocado growth. I label this index specification in the summary statistics as Index 1 in Table 1. Figure 5 displays the geographical location of this paper's sample and the distribution of avocado suitability index values.

[17]Municipalities have varying altitudes. Based on Figure 4, if a municipality has an altitude range only above 2,505 meters then the I classify the municipality has having low suitability. Otherwise, I also consider January temperature and/or the P/T value.

3.3 Mechanism Data

INEGI's 1990 and 2000 General Census of Population and Housing both record the number of primary sector workers at the municipal level. INEGI's primary sector includes workers in agriculture, livestock, forestry, hunting, and fishing. However, their census data after 2000 only records general employment. To test whether the growth of the avocado industry led to increased employment in the primary sector, I take the difference of the primary sector employment in 2000 and 1990 at the municipal level.

The censuses also include data on population, male population, economically active persons, primary education completion, homes with non-dirt flooring, homes with running water, and homes with electricity. I take the difference of all these values between 2000 and 1990 to construct a cross-sectional data set at the municipal level. The sample includes all municipalities in Michoacán and municipalities in states contiguous with Michoacán. I also combine the 1980-2010 average temperature and precipitation data from SMN to control for weather. To ensure that my analysis only compares rural municipalities, I drop all municipalities with a population above 100,000 from the sample. The summary statistics for this data set are in Table A1 of the appendix.

3.4 Data to Test Time Heterogeneity

To test the effects of avocado price variation after the war on drugs began, I use crime data from SESNSP. The data includes municipal-level recordings of property crime, cattle theft, reported threats, homicide by gunfire, rape, highway robbery, and kidnappings. Some observations for these crimes do not exist for certain municipalities in a given year within the data. I omit observations with any of the 7 crime indicators missing from my sample. This is so that I use the same sample when testing all seven crime indicators. The data spans 2011-2017, includes 438 municipios, and 2,316 observations. The summary statistics for this data are in Table A2 of the appendix. [18]

4 Methodology

My empirical strategy uses a difference-in-differences approach. I evaluate whether changes in the price of avocados result in differential effects on homicide in municipalities more agro-climatically suited to growing avocados. Figure 5 shows my avocado suitability measure.

[18] I use SESNSP data from 2011-2017 because they apply the same methodology for that period. Their micro-level crime data is publicly accessible on the following website: https://www.gob.mx/sesnsp/acciones-y-programas/datos-abiertos-de-incidencia-delictiva

The figure displays that the sample has substantial variation in avocado suitability. This helps ensure that the effects of avocado price fluctuations are not driven by one specific geographic area.

An issue with directly using Mexican producer prices is that they may be endogenous to the outcomes of interest. Many factors, such as international trade relations, domestic issues, and economic factors could bias the results. This form of endogeneity leads to reduced magnitudes in the analysis.

To address the concern of endogeneity, I employ an instrumental variables strategy that exploits weather shocks in California's avocado-producing belt. There is a massive trade relationship between the United States and Mexico. In the second half of the 1990s, California produced approximately 95 percent of the avocados grown in the U.S. and 10 percent of the world's production (USDA, 1999a). In 2018, Mexico was the top supplier of avocados to the U.S. with 87 percent of the U.S. market share. From July 2019 to June 2020, Mexico exported 186,958 metric tons of avocado to the U.S.[19]

I focus on three weather instruments that measure all of California's avocado-producing counties represented in the USDA's National Agricultural Statistics Service California Field Office. I use county and monthly data from NOAA to construct weather indices that average conditions among California avocado-producing counties. I first record NOAA's 1900-2000 average April-July temperature, April-July average precipitation, and average minimum January temperature. For each year, I make $USTEMP_t$ with two steps. First, I record the average temperature of the annual California avocado-producing county from April to July. I then take the difference of the annual county temperature average with the 1900-2000 average temperature. I follow the same procedure for $USPREC_t$ and $USMIN_t$, which are U.S. precipitation and January minimum temperature deviations, respectively. To calculate $USPREC_t$, I take the difference of the annual April-July California average precipitation values with the average 1900-2000 April-July precipitation values. For $USMIN_t$, I take the difference of the annual January minimum temperature California average values with the average 1900-2000 January minimum temperature values.[20] I utilize 2-year lags of these three instruments for avocado price year t-1 because the effects of California weather shocks destroying avocado trees have a delayed impact on Mexican producer prices.[21]

[19]To meet U.S. demand, the U.S. imports Mexican avocados year round. When dealing with import contracts, avocado importers should arrange for inspection and certification at least one day prior to entry at Port inspection offices and at least two days prior to entry at the Field Operations Section." (USDA, 2024)

[20]For example, if the 1900-2000 April-July average temperature among avocado-producing municipalities is 75 degrees and the average April-July temperature of avocado-producing counties in a specific year is 80 degrees, then $USTEMP_t$ equals 5 for that given year.

[21]For the the entirety of my analysis, the 2-year lag for the weather instruments are in relation to price.

Let Y_{it} represent log+1 homicide in municipality i and year t. My basic second-stage specification is shown in Equation 2:

$$Y_{i,t} = \beta_0 + \beta_1(Index_i \cdot \widehat{ln(}Price_{t-1}) + \beta_2 X'_{i,t} + \beta_3 Niño_{i,t} + \beta_4 Weather_i + \alpha_{2,i} + \gamma_{2,t} + \varepsilon_{i,t} \quad (2)$$

The coefficient of interest is the interaction term $Index_i * Price_{t-1}$. My preferred measure assesses lagged producer price because people often need time to adjust to their income situations before making choices to commit crime.[22] $X'_{i,t}$ is a vector of INEGI census controls that allow me to account for municipal logged population, logged male population, logged number of economically active persons over the age of 15, and percent of the population that has completed primary education in 1990, 2000, and 2010. I match municipal homicide observations from 1997-1999 with the 1990 census to control for these census variables. In addition, I match municipal homicide observations from 2000-2009 with the 2000 census data and homicide observations from 2010-2019 with the 2010 census data. Niño is a dummy variable that equals 1 if an observation takes place during El Niño, 0 otherwise.[23] Weather is a vector of temperature and precipitation controls at the municipal level. The temperature and precipitation controls are the 1980-2010 annual weather station averages from SMN. $\alpha_{2,i}$ and $\gamma_{2,t}$ represent municipality and year-fixed effects. The first stage equation explaining $Index_i * Price_{t-1}$ is shown in equation 3:

$$\begin{aligned} Index_i \cdot ln(Price)_{t-1} = {} & \beta_0 + \beta_1(Index_i \cdot USTEMP_{t-3}) \\ & + \beta_2(Index_i \cdot USPREC_{t-3}) \\ & + \beta_3(Index_i \cdot USMIN_{t-3}) \\ & + \beta_4 X'_{i,t} + \beta_5 Niño_{i,t} + \beta_6 Weather_i \\ & + \alpha_{1,i} + \gamma_{1,t} + \varepsilon_{i,t} \end{aligned} \quad (3)$$

In equation 3, $\alpha_{1,i}$ and $\gamma_{1,t}$ depict first-stage municipal and year fixed effects, respectively. After planting an avocado tree, it can take 3 to 4 years before yielding fruit. It can take 5 to 13 years after sowing an avocado seed for the tree to mature enough to produce

If price represents t, the weather instruments will be from t-2. If the price represents t-1, then the weather instruments will be from t-3.

[22]Note that in equation 2, I do not include the base terms of the interaction separately. This is because $Price_{t-1}$ is absorbed by year-fixed effects, and $Index_i$ is absorbed by municipality effects.

[23]El Niño years include 1997-1998, 2002-2003, 2004-2005, 2006-2007, 2009-2010, 2014-2016, 2018-2019. (Donegan, 2019; NOAA, 2019; The Independent, 2023)

fruit (California Avocado Commission, 2024). Dube et al. (2016) instruments one year lag of U.S. weather shocks on maize prices because corn is sown and harvested in the same season. In some cases, planted corn can emerge as quickly as 18-21 days after being sown (Ritchie, 1993). Damage to avocado trees will likely have prolonged impacts in later years on Mexican producer prices because of the time it takes to grow avocados.

Figures 6-8 juxtapose standardized logged producer price with t-2 California avocado producing county temperature, precipitation, and January minimum temperature from 1997-2019. Figure 6 shows a close relationship between U.S. average temperature deviations and Mexican producer prices. The figure often shows an opposite trend between the two throughout much of the period. Figures 7 and 8 juxtapose standardized logged producer prices with standardized U.S. precipitation deviations and standardized January minimum deviations, respectively. Again, it is visually apparent that there is a close relationship between producer prices and two-year lagged U.S. precipitation and January minimum temperature deviations. However, both precipitation and January minimum temperature two-year weather lags appear to follow the same type of oscillations throughout the period.

5 Results

5.1 Avocado Prices and Homicide

This section examines the relationship between avocado prices and homicide in Mexico. Table 2 displays the results of equations 2 and 3. All regressions include municipality and year-fixed effects. All robust standard errors are clustered at the municipality level. The response variable is log+1 homicide.[24] Columns 1-4 show the results of the OLS estimations. To test the robustness of the findings, I measure the impact of producer price in year t in columns 1-2 and year t-1 in columns 3-4. The OLS results are statistically significant and consistent regardless of whether I include right-hand controls. The preferred OLS specification is in column 3, where I regress lagged price and include right-hand controls. The coefficient is -0.046 and has a p-value below 0.05. The estimate indicates that a 100 percent increase in price decreases the impact of a unit increase of the suitability index on homicide by approximately 4.6 percent. This magnitude is significant because the producer price range experiences more than a 100 percent increase in the sample period. These results indicate that there is a strong negative association between producer price and homicide in Mexico.

[24] Table A8 in the appendix represents the results of using the inverse hyperbolic sine of homicide instead of the log+1 value. The results are very similar regardless of which way I measure homicide.

The OLS estimations are likely biased due to the endogenous nature of avocado prices. I instrument two-year-lagged U.S. weather shocks in California's avocado-producing counties to address this endogeneity issue. The IV estimates are larger in magnitude than the OLS estimates, consistent with reverse causality from supply effects biasing the least squares estimates toward zero. Columns 5-6 represent measures using price t, and 7-8 show the results from price t-1. The results are consistent and strongly significant with or without right-hand controls, regardless of whether I measure price t or t-1. The p-values for columns 5-8 in the 2SLS specification become much stronger, with values below 0.01. Column 7 shows the preferred 2SLS measure that evaluates the impact of price t-1 on homicide and includes all right-hand controls. The coefficient value of the interaction term is -0.162. The 2SLS magnitude is significantly larger than the OLS and is a result of addressing the endogeneity of the OLS regressions. The 2SLS estimates suggest that a 100 percent increase in price causes a decrease in the impact of a unit increase of the suitability index on homicide by approximately 16.2 percent. The F-statistic has a high value of 1.4e+04. This result is similar to the F-statistics in Dube et al. (2016) and indicates that the instruments are highly relevant to the producer price.[25] The first stage results for Table 2 are in Table A4 of the appendix. Table A4 shows that all instruments are statistically significant for the interaction of the index and producer prices.

The negative relationship between the interaction term and homicide matches the findings of a large body of literature showing commodity price shocks having negative relationships with price (Brückner and Ciccone 2010; Berman and Couttenier 2013; Bazzi and Blattman 2014). The findings of Table 2 support the Becker utility model (Becker, 1968). As people have better employment opportunities and wages, their utility from committing crimes decreases. To test whether this is the case, I utilize data from INEGI's 2000 and 1990 Census to evaluate the difference of primary employment between 2000 and 1990 at the municipal level. [26] The census data allows me to control for the 2000 and 1990 differences between population, economically active persons, primary education completion, homes with flooring, homes with running water, and homes with electricity. I also control for the 1980-2010 municipal average temperature and precipitation from SMN's weather station data. I omit all municipalities with a population above 100,000 so that my data

[25]The F-statistics in my empirical analysis are high. The F-statistic is this high here, and in the literature, because of what happens when clustering robust standard errors in a 2SLS model for an interaction term. Table A7 in the appendix shows the 2SLS results of Table 2, but without clustering robust standard errors at the municipality level. The F-statistics range from 277.44 to 317.64 when I omit robust standard error clusters, which still indicates that my instruments are very relevant to price.

[26]For example, if primary employment in a municipality is 200 jobs in 2000 and 150 jobs in 1990, then the difference would be 50.

only compares among rural regions. The result of amalgamating these variables is a cross-sectional data set that includes 449 municipalities. The benefit of taking the differences between 2000 and 1990 is that NAFTA took place in 1994, and the U.S. opened up the avocado trade with Mexico in 1997. The difference values will likely heavily reflect the impact that avocado trade openness initially had on agricultural employment in Mexico.

I use the following OLS equation to test the association between the avocado suitability index and the difference in primary jobs between 2000 and 1990:

$$Jobs_i = \beta_0 + \beta_1 Index_i + \beta_2 X_i' + \varepsilon_i \tag{4}$$

The response variable represents the difference in primary sector jobs in 2000 and 1990 at the municipal level i. The coefficient of interest is the avocado suitability index to assess whether municipalities more suitable for avocado production experienced an increase in primary employment after avocado trade opened between the U.S. and Mexico. X' is a vector of right-hand census and weather controls at the municipal level.

The results of Equation 4 are in Table 3. All robust standard errors are clustered at the municipality level. Column 1 shows that when I include right-hand controls, the coefficient value for the Index is 38.975 and is statistically significant. This implies that a 1 unit increase in the avocado suitability index is associated with almost 39 additional primary sector jobs between 2000 and 1990. This evidence supports the argument that openness to avocado trade between the U.S. and Mexico increased employment opportunities for Mexico. Based on the Becker utility model, the increased employment opportunities likely resulted in the Mexican people having less utility from committing crime because of their wage increase. Column 2 shows that the results are robust without right-hand controls.

Columns 3-6 in Table 3 test the robustness of the results in columns 1-2. For columns 3-4, I interact the avocado suitability index with the difference in Mexico's national avocado production between 2000 and 1990.[27]. In columns 5-6, I interact the avocado suitability index with the difference in avocado production value between 2000 and 1990 in pesos and real terms. The results remain consistent and robust for each interaction term with or without right-hand controls.

[27]Data for Mexico's national output of avocado and production value in 2000 and 1990 come from Mexico's Servicio de Información Agroalimentaria y Pesquera (SIAP) and is publicly available on the following website: https://nube.siap.gob.mx/cierreagricola/

5.2 Robustness Checks

The results of Table 2 hold to a vast range of robustness checks. In addition to testing the impact of producer prices on homicide, I also evaluate the relationship of FAS converted and non-FAS converted avocado import prices from Mexico to the U.S. Table 4 displays the impact of both import prices on homicide. Columns 1-4 show the OLS and 2SLS results when analyzing non-FAS converted import prices. The results are all statistically significant and consistent. The coefficient value of the interaction term in column 1 is -0.110, which is more than double the magnitude of the OLS results with producer price. In addition, the p-values are lower when measuring non-FAS import prices than when measuring producer prices. The p-values are below 0.01 for all the results in columns 1-4. Column 3 shows the 2SLS results with right-hand controls. The coefficient value of the interaction term in this case is -0.165, which is almost the same magnitude as the 2SLS method with producer price. The F-statistic is 2.1e+05. This F-statistic is larger for the non-FAS import price than the producer price, which implies that the U.S. weather shocks may have a bigger effect on non-FAS converted import prices than the producer price. The results remain significant and consistent with or without controls.

Columns 5-8 of Table 4 show the impact of lagged FAS-converted import prices on homicide. The OLS and 2SLS results are statistically significant. The OLS magnitudes are similar to the non-FAS import price results; however, the 2SLS magnitudes are smaller. Column 7 shows the 2SLS results with right-hand controls. Unlike the two other price measures, the magnitude of the interaction term's coefficient is smaller when using the 2SLS model than the OLS method. The F-statistic in this case is 1.0e+05, which is bigger than the F-statistic values for the regressions assessing producer price. All results remain significant and consistent with or without right-hand controls. In short, all specifications of Table 4 show a negative relationship between the interaction of the index with lagged avocado prices and homicide in Mexico.

One concern in the analysis is whether the results mostly come from the state of Michoacán. From 1997-2019, Michoacán was the only state permitted to export their avocados to the U.S. To assuage these concerns, I test the OLS and 2SLS results for all 3 lagged price measures when I omit Michoacán from the sample. Table 5 shows the results without Michoacán. Both the OLS and 2SLS results for all 3 lagged price measures are statistically significant and consistent. Interestingly, the 2SLS magnitudes for all 3 price measures in Table 6 show larger coefficient values when Michoacán is not in the sample. For example, column 2 shows the 2SLS results when measuring lagged producer prices. The magnitude of the interaction term is -0.225. This finding indicates that a 100 percent increase in price decreases the impact of a unit increase of the suitability index on homicide by approximately 22.5 percent. That is roughly 6 percent higher than the magnitude of the interaction term when Michoacán is in the sample. A possible explanation of this is that Michoacán is a war-torn state from the territorial contestation between La Familia Michoacána and other drug cartels in the region. Including a severely war-torn state would thus lower the estimates of a negative relationship between avocado prices and homicide.

I further show that my results hold to multiple index specifications. Equation 6 shows an alternative index calculation I test:

$$Index_i = \frac{S_i + C_i}{2} \tag{5}$$

Instead of taking the sum of the soil and climate values, I take their average. Table 6 shows the OLS and 2SLS results of the impact of the interaction term on homicide for all three lagged price measures. The results show larger magnitudes for all OLS and 2SLS results among each lagged measure. For example, the coefficient value of the 2SLS specification in column 2 is -0.353. This suggests that a 100 percent increase in producer price decreases the impact of a unit increase of the suitability index on homicide by approximately 35.3 percent. The 2SLS coefficient for the interaction term with non-FAS converted avocado import price and FAS converted avocado import price is -0.437 and -0.409, respectively. This suggests that a 100 percent increase in import price decreases the impact of a unit increase of the suitability index on homicide by approximately 40.6 to 43.7 percent. The results for all OLS and 2SLS methods and lagged price measures are statistically significant and consistent. I also test my results with a third index measure. Instead of the avocado suitability index equaling the sum of the soil and climate values and 0 if either of the two values are 0, I apply the sum of the the soil and climate values. For example, if the index value of S_i=3 and $C_i = 0$, the index value would equal 3 (it would equal 0 in the original index's specification). The results when I use this third index are in Table A9 of the appendix. The findings of the analysis with this third index classification remain statistically significant and consistent. The summary statistics for the second and third indexes are in Table 1.

To test the robustness of the 2SLS method in my analysis, I explore using different instruments. The results of applying alternative instruments are in Table 7. Weather variables can often take on non-linear patterns.[28] Column 1 shows the results of when I square my three weather instruments. The results are statistically significant, with a

[28]Dube et al. (2016) also explores squared U.S. weather shock IVs as instruments for maize price.

coefficient of -0.165, which is similar to the magnitude of the original weather instruments. The F-statistic is 5.1e+03, which is smaller than before but still large. In column 2, I include both the weather instruments and squared instruments. The results are statistically significant; however, the magnitude of the coefficient is -0.038, which is much smaller than the results of the previous 2SLS methods.

Nunn and Qian (2014) uses a U.S. lagged wheat price interaction term on foreign aid to assess how foreign aid impacts civil conflict in recipient countries. Based on their method, I test my model using an alternative IV strategy that interacts lagged producer price ($Price_{t-2}$) with my suitability index. The result is in column 3 of Table 5. The coefficient is -0.065 and is statistically significant, with a p-value below 0.01. In addition, the F-statistic is 2.5e+04, which shows that this is a strong instrument. Column 4 shows that the results hold when I add the lagged producer price interaction instrument with the original 3 weather instruments of my model. Column 5 displays the results of adding all seven instruments in the 2SLS model. The coefficient is -0.06 and is significant. The F-statistic is 2.0e+05, much larger than the results of columns 1-4. The reformed first stage equation for column five of Table 7 is the following:

$$\begin{aligned}
Index_i \cdot ln(Price)_{t-1} = \beta_0 &+ \beta_1(Index_i \cdot USTEMP_{t-3}) \\
&+ \beta_2(Index_i \cdot USPREC_{t-3}) \\
&+ \beta_3(Index_i \cdot USMIN_{t-3}) \\
&+ \beta_4(Index_i \cdot USTEMP^2_{t-3}) \\
&+ \beta_5(Index_i \cdot USPREC^2_{t-3}) \\
&+ \beta_6(Index_i \cdot USMIN^2_{t-3}) \\
&+ \beta_7(Index_i \cdot ln(Price)_{t-2}) \\
&+ \beta_8 X'_{i,t} + \beta_9 Ni\tilde{n}o_{i,t} + \beta_{10} Weather_i \\
&+ \alpha_{1,i} + \gamma_{1,t} + \varepsilon_{i,t}
\end{aligned} \tag{6}$$

Table A5 in the appendix shows the first-stage results of equation 6. As additional IV robustness checks, I test different combinations of my original 3 IVs. My results remain strong and consistent when applying single weather instruments or many combinations of the 3. These results are in Table A6 in the appendix.

To look deeper into the lagged effects of U.S. weather shocks on avocado producer and import prices, I evaluate equation 3 with U.S. weather shocks from different periods. Table 8 displays the results of different U.S. weather lags. Equation 3 assesses lagged price, and my preferred method shows the impact of 2-year lagged U.S. weather shocks, which are in t-3. Thus, Table 8 does not show the results for t-3. Columns 1-3 represent the results for different U.S. weather shock years on lagged producer prices. Columns 1 and 2 show no statistical significance when using weather instruments from the same year of the producer price or 1-year-lagged weather instruments. However, column 3 shows that the results are statistically significant when I use 3-year lagged U.S. weather shock instruments. The coefficient value is -0.136, slightly smaller than the magnitude of the interaction term from 2-year lagged weather instruments. This indicates that U.S. weather shocks on avocado trees have lagged effects on producer prices. In other words, U.S. weather shocks from 2 or more years seem to impact producer prices significantly, but the biggest effect on price comes from 2-year lags.

Columns 3-6 in Table 8 show the 2SLS results of different weather instrument periods with import prices. The results are statistically significant for all periods. This finding indicates that U.S. weather shocks impact import price in the same year. In addition, U.S. weather shocks from 1 year and 3 years before the year of price are significant to the results. In short, these findings indicate that the impact of U.S. weather shocks impacts import prices quicker than they do producer prices.

5.3 Time Heterogeneity

In 2006, Mexico's government began the war on drugs. Mexico's government started aggressively cracking down and fighting drug cartels. This sparked a horrifying backlash from drug cartels that led to increased levels of crime. To fund territorial contestation, drug cartels began diversifying their portfolios by taking over export agricultural products such as avocados. I next test whether the relationship between the interaction term of the index and avocado producer prices and crime remain negative from 2006 to 2019.

Table 9 shows the results of Table 2 when I omit the years before 2006 from the sample. The sample contains 6,192 observations, and the relationship between avocado prices and crime changes. All 8 columns show a positive and statistically significant relationship between avocado producer price and homicide. The preferred OLS measure is in column 3, where I evaluate lagged producer price and include right-hand controls. The coefficient value is 0.087. This indicates that a 100 percent increase in producer price increases the impact of a unit increase of the suitability index on homicide by approximately 8.7 percent. This is almost double the magnitude of the OLS estimate when all the years of the data are included, but with the opposite sign. Column 7 displays the preferred 2SLS measure, where I examine the impact of lagged producer price on homicide. The coefficient value of 0.741 is significant, with a p-value close to 0. The F-statistic remains high at 8.4e+05. This turn of events suggests that a 100 percent increase in price increases the impact of a unit increase of the suitability index on homicide by approximately 74.1 percent. This magnitude is significantly larger than the same method's results with all of the years in the sample, but with the opposite sign. These results provide strong evidence that after the 2006 war on drugs began, avocado producer price increases led to more homicide.

The heterogeneous change in the relationship between avocado prices and homicide matches the literature showing how commodity price shocks can lead to increased crime. The mechanism behind this change is the war on drugs, which made cartels have to more aggressively diversify their portfolios to the avocado industry to fund their fighting. A large body of literature discusses the growing concern of drug cartel presence among licit commodity production (García-Ponce and Lajous, 2014; Bergman, 2018; Nelleman et al., 2016; Zabyelina and van Uhm, 2020; Calderón et al., 2020; Felbab-Brown, 2019; Guerrero Gutiérrez, 2011; 2020; Durán-Martínez, 2018; Flores-Macías, 2018). Magaloni et al. (2020) highlights the War on Drugs in Mexico intensified competition among cartels, which incentivized drug trafficking organizations to expand their revenue-generating activities to new economic sectors besides illicit commodities. A significant body of work supports the fact that drug trafficking and political backlash against drug cartels have led to massive increases in crime (Dell et al., 2015; Leenen and Cervantes, 2014; Gamlin, 2015). Thus, criminal organizations in Mexico require secure sources of income to build up their violence-making capabilities as they face fierce competition and state crackdowns (Correa-Cabrera, 2017; Jones, 2016).

I explore the relationship of avocado producer prices with more crime categories using data from SESNSP, which includes municipal and annual observations for property crime, cattle theft, reported threats, homicide by gunfire, rape, highway robberies, and kidnapping from 2011 - 2017. I control for El Niño and the municipality population and utilize both OLS and 2SLS methods. The results are in Tables 10 and 11. Columns 1 and 2 display the relationship between avocado producer prices and property crime. The coefficients are 0.162 and 0.208 for the OLS and 2SLS models, respectively. These statistically significant results show a positive relationship between property crime and avocado producer prices.[29] Reports have stated drug cartels often steal Mexican farmers' land and begin planting avocado trees (Romero and Mega, 2023). Such reports discuss how when farmers ask the cartels to leave their land, they receive threats toward them and/or their families from the

[29]Each column measures the log+1 value of the response variable. Table A10 in the appendix shows the results of Table 10 when I measure the inverse hyperbolic sine of each response variable instead. The results hold regardless of which way I measure the response variables.

cartels.

Columns 3 and 4 of Table 10 show the OLS and 2SLS methods measuring the relationship between cattle theft and the interaction term. The OLS and 2SLS coefficients are 0.157 and 0.217, respectively, and are statistically significant. The positive relationship between cattle theft and producer prices intuitively makes sense. If drug cartels are brazen enough to steal people's land openly, then it is highly likely they would be willing to steal their cattle.

The response variable of columns 5 and 6 is reported threats, while the response variable for columns 7 and 8 is homicide by gunfire. The signs of the results are all positive and statistically significant, which indicates avocado producer price increases lead to increased reported threats and homicide by gunfire. The increase in reported threats matches reports of cartels threatening farmers to give them their land and/or pay protection fees for their avocado production. The results of homicide by gunfire are consistent with the homicide results of 2006 to 2019. The 2SLS results for each response variable have larger magnitudes, which is consistent with reverse causality from supply effects biasing the least squares estimates toward zero. To test the robustness of the results in Table 10, I evaluate the 2SLS methods with alternative instruments for either squared weather outcomes or lagged producer prices. The results are strongly robust among these alternative instruments and are in Table A11 in the appendix.

Table 11 displays the OLS and 2SLS results of measuring the relationship between avocado prices and rape, highway robbery, and kidnappings. While the sign of almost every regression in Table 11 is positive, none of the results are statistically significant. The lack of statistical significance between producer price and either rape or highway robberies intuitively makes sense as these are not crimes closely related to agriculture. However, the lack of significance in the relationship between avocado producer prices and kidnappings is somewhat surprising. Media outlets such as the New York Times have reported victims discussing how they were kidnapped and beaten due to disputes between drug cartels and Mexican farmers. While the results of kidnapping in columns 5 and 6 may be true, this crime data is limited to the amount of abductions people report. Likely, many victims do not report kidnapping cases due to fear of what drug cartels will do.

6 Conclusion

Numerous media outlets have discussed the growing concern of Mexico's avocado production and cartel involvement (Steel and Ember, 2023; Swanson, 2022; Flannery, 2023; NPR, 2023; The Economist, 2016). A growing concern among politicians and the general public is whether the massive growth of the avocado industry and its involvement with drug cartels has led to more crime in Mexico. Many news outlets have asked harrowing questions, such as whether U.S. avocado demand is actively funding drug cartels. This paper creates a novel municipality-level avocado suitability index using agricultural science methods in order to assess a causal mechanism between avocado prices and crime in Mexico. This article is the first to rigorously explore the causal relationship between avocado prices and crime in Mexico. This is also the first paper to provide empirical evidence that lagged U.S. weather shocks in California's avocado-producing counties impact crime in Mexico through avocado prices.

This paper offers empirical evidence showing the gray relationship between avocado prices and crime in Mexico. By creating a novel avocado suitability index at the municipal level, I evaluate whether changes in the price of avocados result in differential effects on homicide in municipalities more agro-climatically suited to growing avocados. When assessing the period 1997-2019, both my OLS and 2SLS methods show that avocado prices have a negative relationship with homicide. My estimates suggest that a 100 percent increase in avocado producer prices decreases the impact of a unit increase of the avocado suitability index on homicide by approximately 16 percent. This is a significant magnitude as the Mexican avocado producer price increases by more than 100 percent over this period. These findings support the Becker utility model that as people have more employment opportunities and higher wages they receive less utility from committing crimes. My analysis empirically shows that avocado trade openness between the U.S. and Mexico is strongly associated with increased municipality-level primary sector jobs.

On the other hand, this paper shows that after Mexico's war on drugs began in 2006, increases in avocado producer prices led to increased homicide, property crime, cattle theft, and reported threats in Mexico. These results are a sobering indication that since 2006, drug cartels have funded their territorial contestation and violence by diversifying their economic portfolios to include the avocado industry. Unfortunately, cartel violence in this period seems to have a bigger impact on crime than the increased employment and wage opportunities of the avocado industry. These findings are essential for U.S. and Mexican policymakers to be aware of as trade with the two countries continues to grow and they both attempt to find a way to address the instability of drug cartel control in Mexico.

Understanding how to address the drug cartel conflict in Mexico is a highly complex task. The answer is not as easy as stopping trade between the two countries or specifically stopping the purchase of avocados, as this will only lead drug cartels to focus on other commodities. Future research should continue to focus on how commodity price shocks from trade impact crime among different nations. Drug cartels in Latin America are heavily involved in numerous industries such as iron, ore, and many other export agricultural products. It is important to know how these products' trade relationships and price variations between different countries may or may not impact crime. Sadly, this paper finds that the delicious guacamole so many enjoy comes at more than just a monetary cost.

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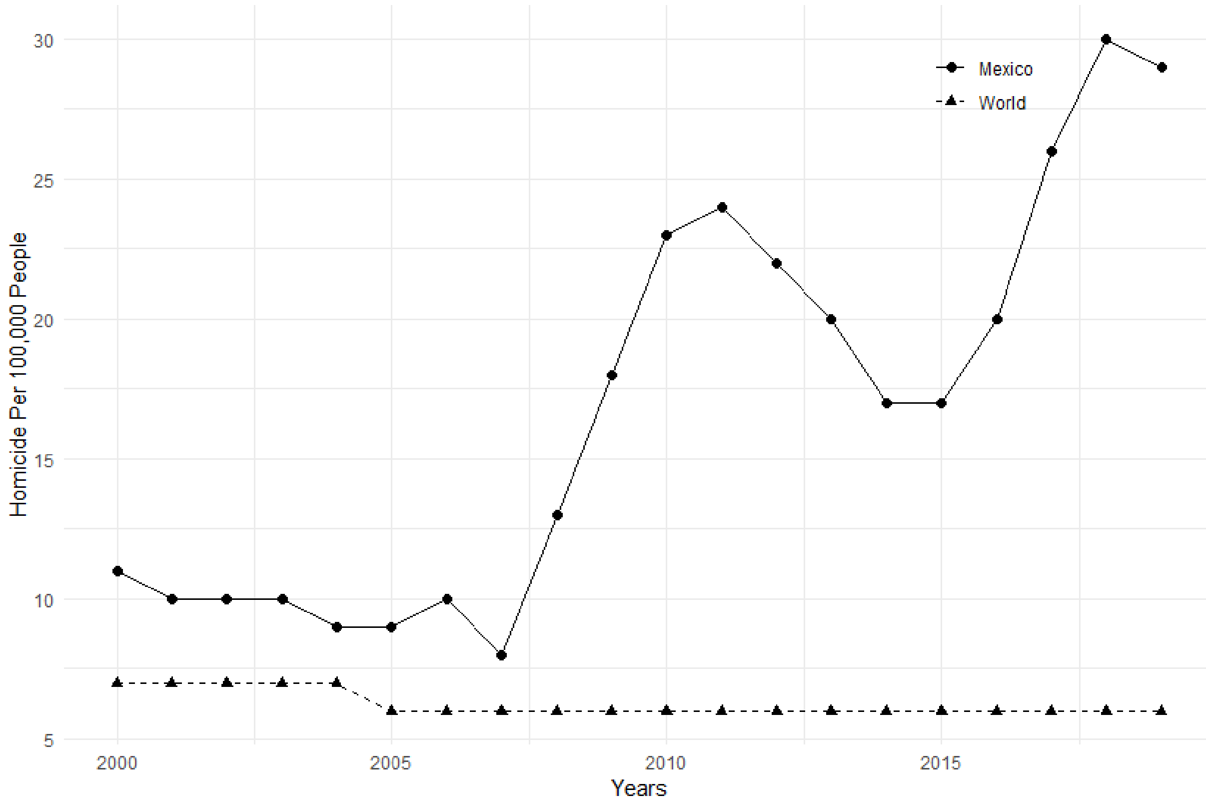
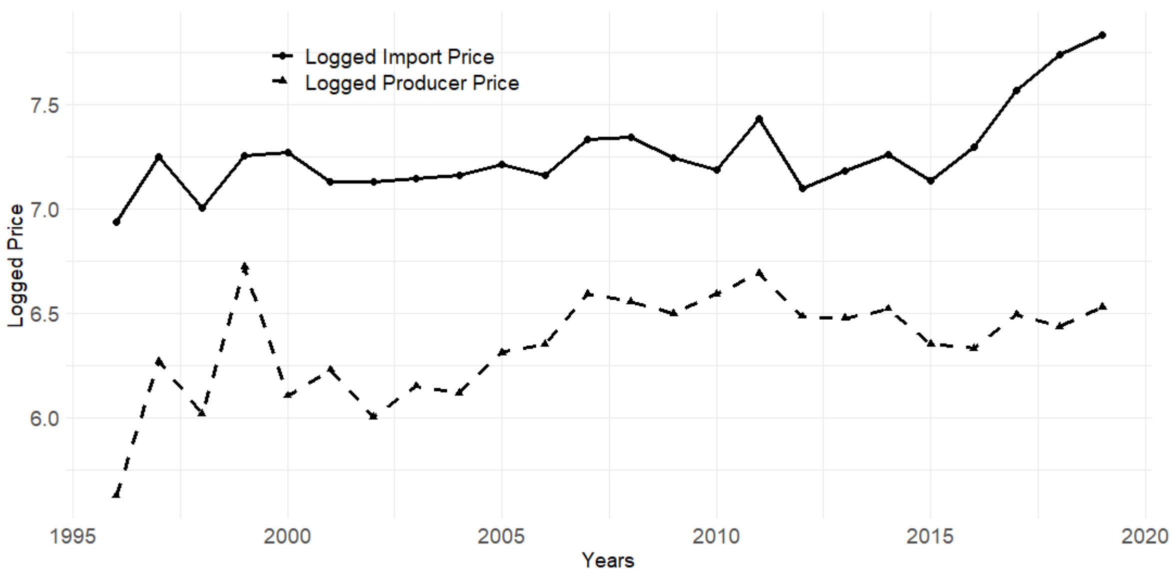
7 Figures and Tables

Figure 1: Homicide Per 100,000 People from 2000 to 2019



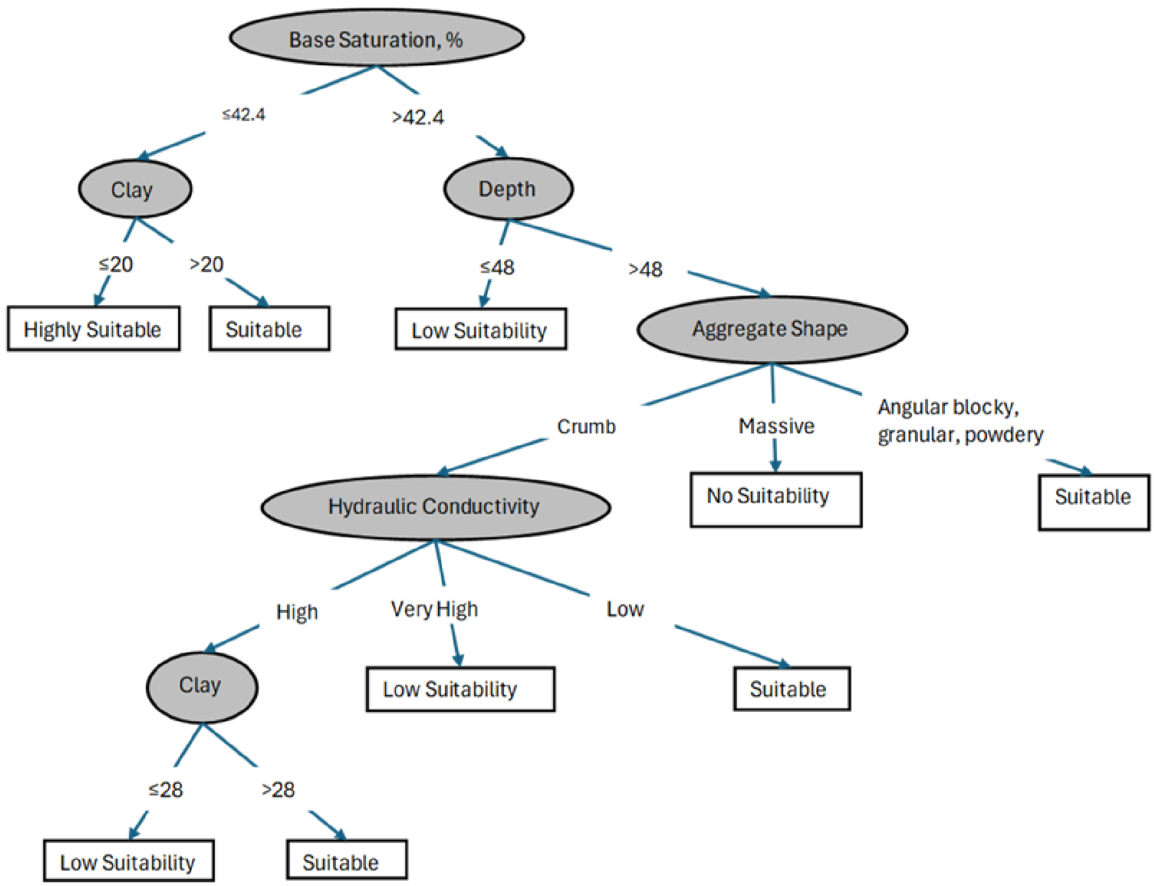
This figure shows the WBO homicide rate per 100,000 people from 2000 to 2019. The solid line with circles represents Mexico's homicide rates over time. The dashed line with triangles shows the world homicide rates over time.

Figure 2: Avocado Producer and Import Prices Over Time



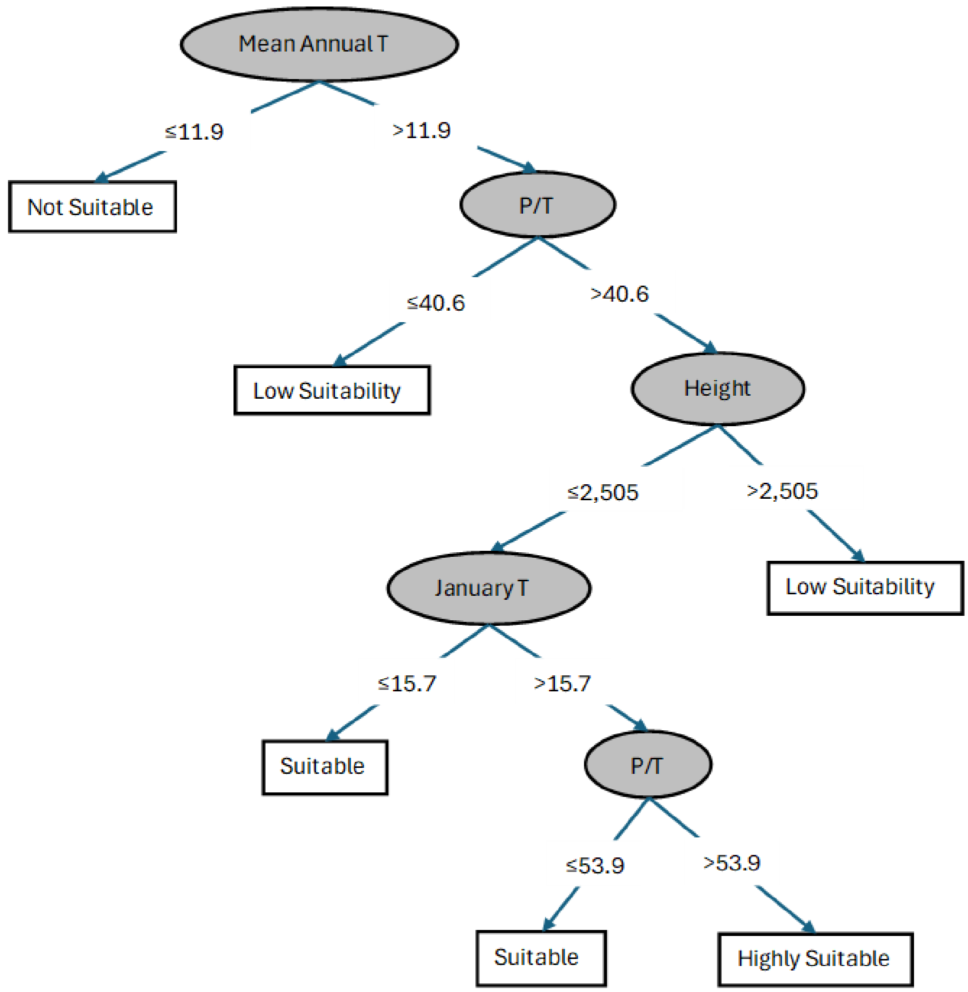
The solid line with circles represents logged real non-FAS import price in USD/Ton from GATS data. The dashed line with triangles shows logged real Mexican producer prices in USD/Ton from FAO data.

Figure 3: Avocado Soil Suitability Tree



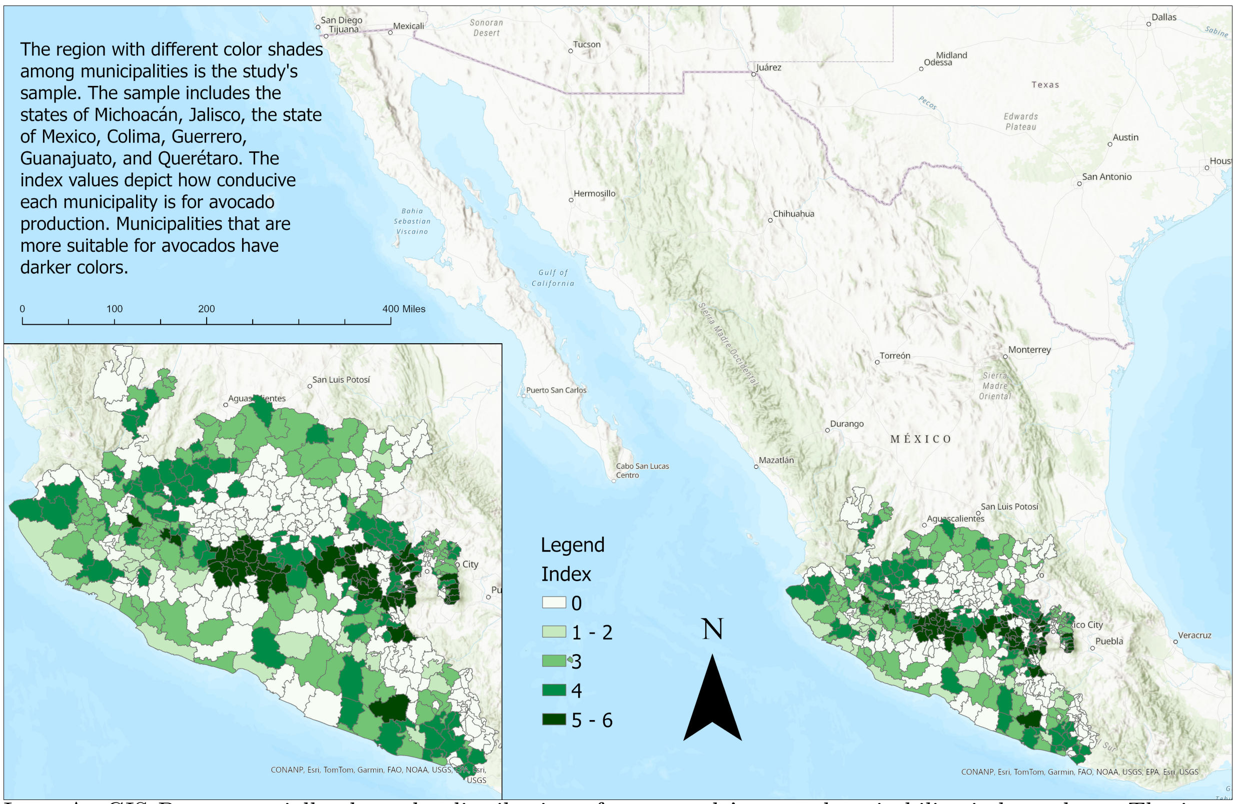
This image represents the avocado soil suitability methodology from Dubrovina and Bautista (2014). Base saturation and clay are in percentage terms, and depth is in centimeters.

Figure 4: Avocado Climate Suitability Tree



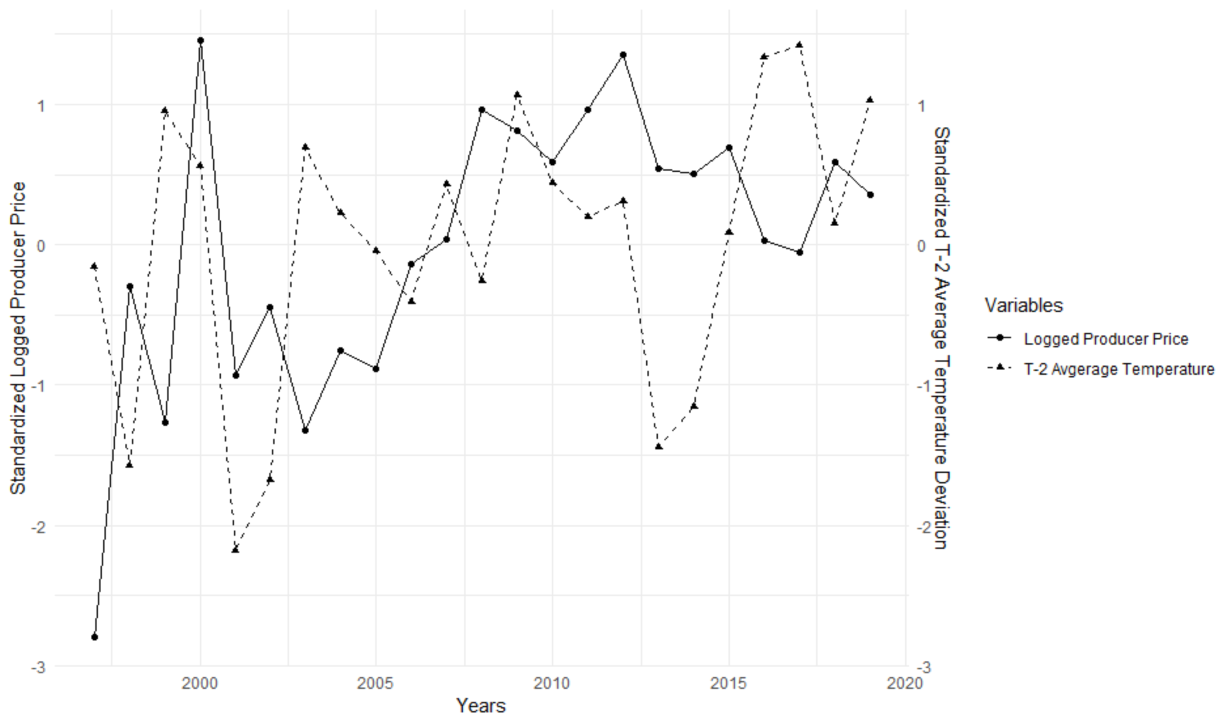
This image represents the avocado soil suitability methodology from Dubrovina and Bautista (2014). Mean annual t is in Celsius, P/T is the ratio of precipitation in inches divided by mean annual temperature, height represents altitude in meters, and January stands for the average temperature in January.

Figure 5: Map of Avocado Index



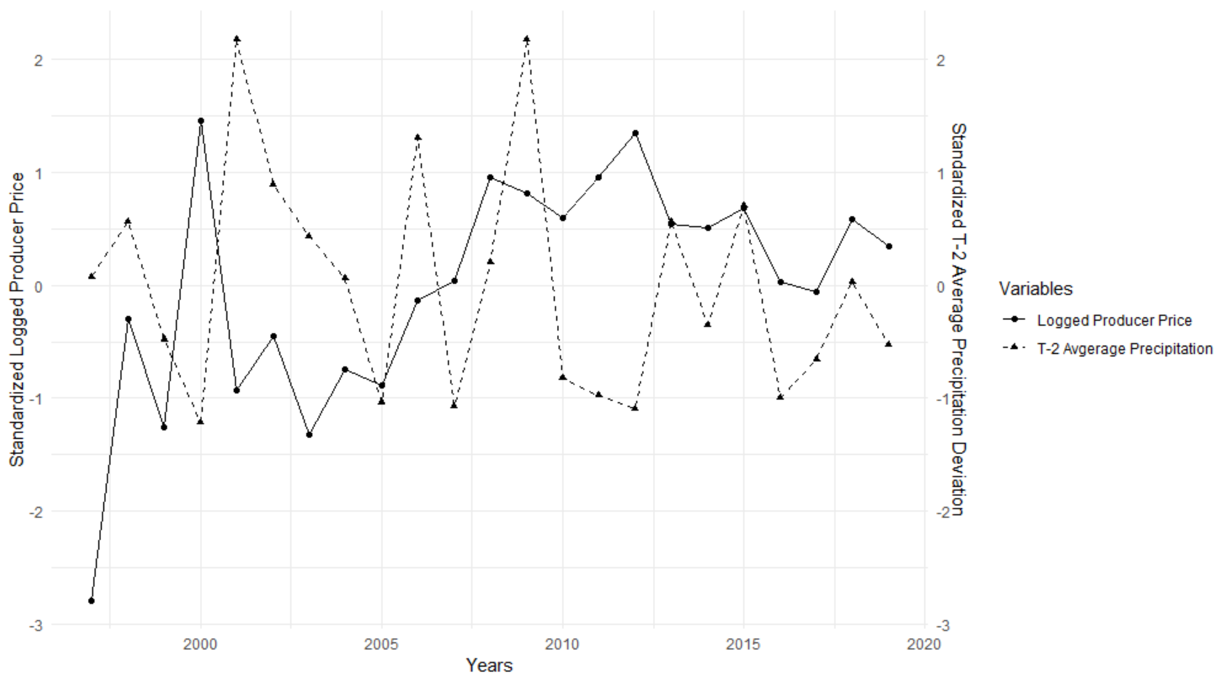
I use ArcGIS Pro to spatially show the distribution of my sample's avocado suitability index values. The inset map at the bottom left shows a zoomed-in map of the sample. The sample includes Michoacán and all of its bordering states. White represents municipalities that are not suitable for avocado trees, while the darker the green in a municipality the more suitable it is for avocados. The darkest green spot in the center is Michoacán's avocado-producing belt.

Figure 6: Producer Price and T-2 Average Temperature Deviation Comparison



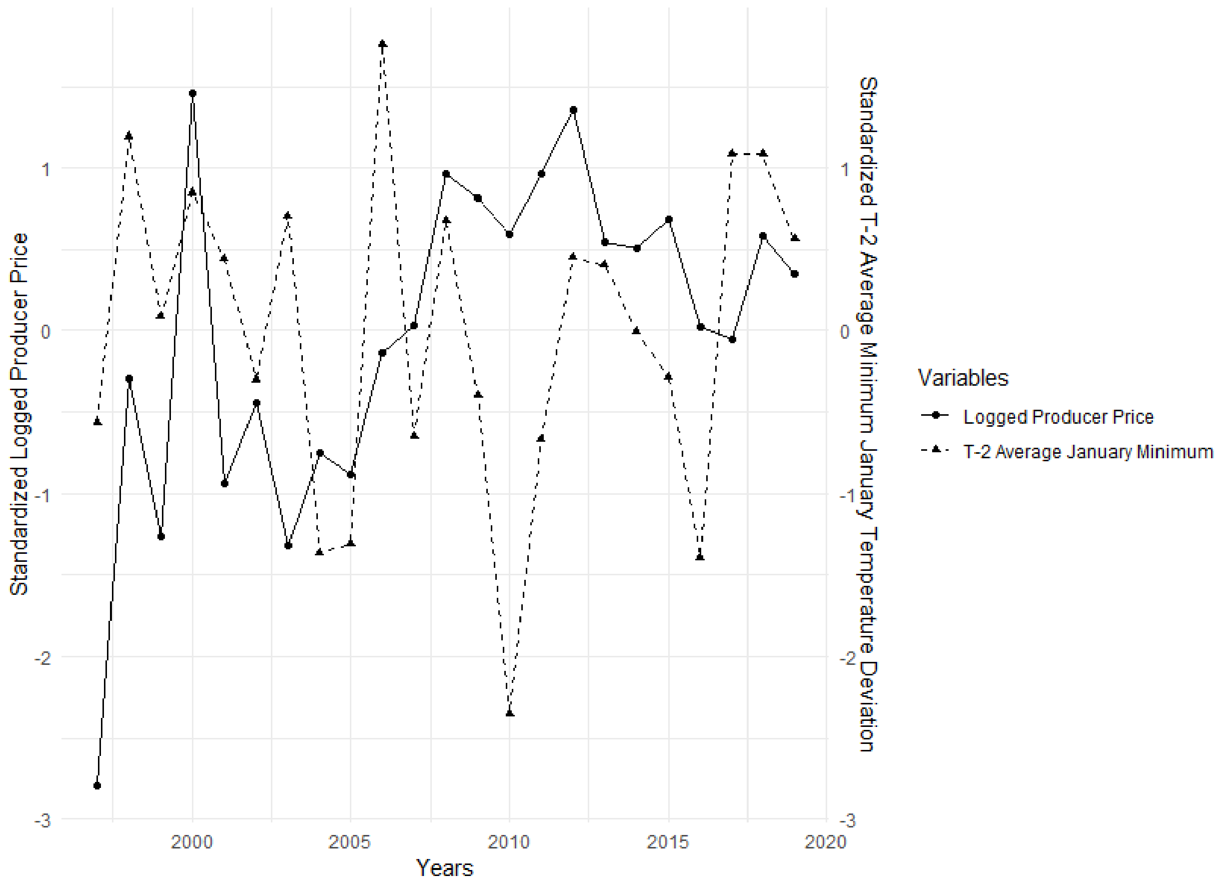
The solid line with circles represents the standardized logged real Mexican producer price (USD/Ton) from 1997 to 2019. The dashed line with triangles shows the standardized two-year lag of the average temperature deviation (in degrees) from producer price among California's avocado-producing region. To calculate the U.S. weather deviations, I take the difference of a year's California avocado-producing county's average temperature from April-July with NOAA's 1900-2000 mean temperature from April-July. California's county average temperature data comes from NCEI and producer prices come from FAO.

Figure 7: Producer Price and T-2 Average Precipitation Deviation Comparison



The solid line with circles represents the standardized logged real Mexican producer price (USD/Ton) from 1997 to 2019. The dashed line with triangles shows the standardized two-year lag of the average precipitation (in inches) deviation from producer price among California's avocado-producing region. To calculate the U.S. precipitation deviations, I take the difference of a year's California avocado-producing county's average precipitation from April-July with NOAA's 1900-2000 mean temperature from April-July. Data for California county precipitation comes from NCEI and producer prices come from FAO.

Figure 8: Producer Price and T-2 Average Minimum January Temperature Deviation Comparison



The solid line with circles represents the standardized logged real Mexican producer price (USD/Ton) from 1997 to 2019. The dashed line with triangles shows the standardized two-year lag of the average annual minimum temperature in January (in degrees) deviation from producer price among California's avocado-producing counties. To calculate the U.S. average minimum January temperature deviations, I take the difference of a year's California avocado-producing county's average January minimum temperature with NOAA's 1900-2000 mean average January minimum temperature. Data for California county average minimum January temperature comes from NCEI and producer prices come from FAO.

Table 1: Summary Statistics

Variable	Observations	Mean	Std. Dev.	Min	Max
Panel-level municipal variables					
Homicides	10,285	3.940	8.208	0	155
Homicide Rate Per 1,000 People	10,285	0.147	0.243	0	3.898
El Niño	10,285	0.352	0.478	0	1
Population (in thousands)	1,380	26,950.76	21,508.45	2,082	99,870
Logged Male Population (in thousands)	1,380	13,083.17	10,431.63	1,045	49,666
Economically Active Persons	1,380	8,507.998	7,517.544	463	40,277
Primary Education Completion	1,380	3,547.56	2,875.09	218	14,609
Cross-sectional municipal variables					
Index 1	460	2.338	1.933	0	6
Index 2	460	1.439	0.681	0.5	3
Index 3	460	2.877	1.363	1	6
Mexico 1980 - 2010 Temperature Avg. (Celcius)	460	19.620	4.050	10	29.5
Mexico 1980 - 2010 Precipitation Avg. (Celcius)	460	874.144	272.170	302.8	2,267.9
Annual-level variables					
Real Mexico Producer Price (USD/Ton)	23	603.258	118.463	404.940	828.262
Real FAS Converted Import Price (USD/KG)	23	1.387	0.174	1.059	1.920
Real Not FAS Converted Import Price (USD/MT)	23	1,471.068	330.14	1,100.321	2,512
Lagged Avg. Temp. Deviation (Degrees)	23	1.83	1.547	-1.61	4.085
Lagged Avg. Prec. Deviation (Inches)	23	-0.022	1.050	-1.33	2.31
Lagged Avg. Minimum Temp Deviation(Degrees)	23	2.491	2.259	-2.9	6.53

Note: See data section for definitions of variables.

Table 2: OLS and 2SLS Results

	(1) Homicide	(2) Homicide	(3) Homicide	(4) Homicide	(5) Homicide	(6) Homicide	(7) Homicide	(8) Homicide
$\text{Index}_i \cdot$ Ln(ProducerPrice)	-0.046**	-0.041*	-0.046**	-0.033*	-0.185***	-0.197***	-0.162***	-0.160***
	(0.021)	(0.022)	(0.018)	(0.018)	(0.067)	(0.067)	(0.056)	(0.057)
Estimation Method	OLS	OLS	OLS	OLS	2SLS	2SLS	2SLS	2SLS
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Right Hand Controls	Yes	No	Yes	No	Yes	No	Yes	No
F-Stat	-	-	-	-	2.3e+04	5.3e+04	1.4e+04	5.4e+04
Price Year	t	t	t-1	t-1	t	t	t-1	t-1
N	10,285	10,285	10,285	10,285	10,285	10,285	10,285	10,285
R^2	0.603	0.599	0.603	0.599	0.600	0.596	0.600	0.595

Note: Robust standard errors clustered at the municipal level are shown in parentheses. The response variable is the log+1 value of homicide. Variables not shown and included in regressions with right-hand controls include log population, log male population, percent of the population with primary education completion, logged population that is 15 years or older and economically active, the cross-sectional 1980-2010 municipal temperature and precipitation, and a dummy accounting for what years El Niño takes place. Columns 1-2 and 5-6 evaluate producer price at time t, while columns 3-4 and 7-8 measure producer price at time t-1. The F-statistic in each 2SLS regression refers to the Angrist-Pischke F-statistic of excluded instruments. In columns 5-8, the interaction term of the avocado suitability index and logged real producer price is instrumented with the interaction of the avocado suitability index and two-year lagged (in relation to price) weather conditions in the U.S. (average temperature, precipitation, and minimum January temperature). *** is significant at the 1 percent level, ** is significant at the 5 percent level, and * is significant at the 10 percent level.

Table 3: Mechanism

	(1) Primary Emp.	(2) Primary Emp.	(3) Primary Emp.	(4) Primary Emp.	(5) Primary Emp	(6) Primary Emp.
Index	38.975***	60.377***	-	-	-	-
	(11.803)	(15.672)	-	-	-	-
Index · Ln(Quantity)	-	-	4.112***	6.369***	-	-
	-	-	(1.245)	(1.653)	-	-
Index · Ln(ProductionValue)	-	-	-	-	2.671***	4.137***
	-	-	-	-	(0.809)	(1.074)
Right Hand Controls	Yes	No	Yes	No	Yes	No
R^2	0.472	0.035	0.472	0.035	0.472	0.035
N	449	449	449	449	449	449

Note: Robust standard errors clustered at the municipal level are shown in parentheses. Variables not shown and included in regressions with right-hand controls include Mexico's municipal level difference of the 2000 and 1990 INEGI census data for population, male population, economically active persons, privary education completion, homes with non-dirt flooring, homes with running water, and homes with electricity. Additional right-hand controls not shown include the 1980-2010 average municipal level temperature and precipitation. Columns 1 and 2 evaluate the association of the avocado suitability index with the change in primary employment from 1990 to 2000. Columns 3 and 4 show the correlation between the interaction term of the avocado suitability index and the difference of logged total Mexican avocado production in tons from 1990 to 2000 with primary employment change. Columns 5-6 display the association between the interaction term of the avocado suitability index and the difference of logged total Mexico avocado production value in pesos from 1990 to 2000 with primary employment change.

Table 4: Results with Different Avocado Price Indicators

	(1) Homicide	(2) Homicide	(3) Homicide	(4) Homicide	(5) Homicide	(6) Homicide	(7) Homicide	(8) Homicide
Index · Ln(ImportPrice)$_{t-1}$	-0.110***	-0.096***	-0.165***	-0.159***	-0.122***	-0.102***	-0.097**	-0.089**
	(0.032)	(0.033)	(0.039)	(0.039)	(0.032)	(0.033)	(0.039)	(0.039)
Estimation Method	OLS	OLS	2SLS	2SLS	OLS	OLS	2SLS	2SLS
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Right Hand Controls	Yes	No	Yes	No	Yes	No	Yes	No
FAS Converted	No	No	No	No	Yes	Yes	Yes	Yes
F-Stat	-	-	2.1e+05	3.7e+05	-	-	1.0e+05	2.0e+05
N	10,285	10,285	10,285	10,285	10,285	10,285	10,285	10,285
R^2	0.604	0.600	0.604	0.600	0.604	0.600	0.604	0.600

Note: Robust standard errors clustered at the municipal level are shown in parentheses. Variables not shown and included in regressions with right-hand controls include log population, log male population, percent of the population with primary education completion, logged population that is 15 years or older and economically active, the cross-sectional 1980-2010 municipal temperature and precipitation, and a dummy accounting for what years El Niño takes place. Columns 1-4 display the OLS and 2SLS results for lagged log non-FAS converted import prices. Columns 5-8 show the OLS and 2SLS results for lagged log FAS converted import prices. For the 2SLS regressions, the interaction term of the avocado suitability index and logged lag real import price is instrumented with the interaction of the avocado suitability index and two-year lagged (in relation to price) weather conditions in the U.S. (average temperature, precipitation, and minimum January temperature). *** is significant at the 1 percent level, ** is significant at the 5 percent level, and * is significant at the 10 percent level.

Table 5: Results without Michoacán

	(1) Homicide	(2) Homicide	(3) Homicide	(4) Homicide	(5) Homicide	(6) Homicide
Index · Ln(Price)$_{t-1}$	-0.047**	-0.225***	-0.134***	-0.246***	-0.140***	-0.257***
	(0.022)	(0.064)	(0.039)	(0.061)	(0.041)	(0.078)
Estimation Method	OLS	2SLS	OLS	2SLS	OLS	2SLS
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Right Hand Controls	Yes	Yes	Yes	Yes	Yes	Yes
Producer Price	Yes	Yes	No	No	No	No
Not FAS Converted Import Price	No	No	Yes	Yes	No	No
FAS Converted Import Price	No	No	No	No	Yes	Yes
F-Stat	-	1.0e+04	-	2.0e+04	-	1.5e+04
N	7,839	7,839	7,839	7,839	7,839	7,839
R^2	0.623	0.617	0.625	0.624	0.624	0.623

Note: Robust standard errors clustered at the municipal level are shown in parentheses. This table shows results for when Michoacán is not included in the sample. The response variable is the log+1 value of homicide. Right hand variables not shown include log population, log male population, percent of the population with primary education completion, logged population that is 15 years or older and economically active, the cross-sectional 1980-2010 municipal temperature and precipitation, and a dummy accounting for what years El Niño takes place. Columns 1-2 display the OLS and 2SLS results for lagged log real producer prices, respectively. Columns 3-4 show the OLS and 2SLS results for lagged log real non-FAS converted import prices, respectively. Columns 5-6 display the OLS and 2SLS results for lagged log real FAS converted import prices, respectively. For the 2SLS regressions in columns 2, 4, and 6, the interaction term of the avocado suitability index and logged lag real import price is instrumented with the interaction of the avocado suitability index and two-year lagged (in relation to price) weather conditions in the U.S. (average temperature, precipitation, and minimum January temperature). *** is significant at the 1 percent level, ** is significant at the 5 percent level, and * is significant at the 10 percent level.

Table 6: Index Robustness Check

	(1) Homicide	(2) Homicide	(3) Homicide	(4) Homicide	(5) Homicide	(6) Homicide
Index · Ln(Price)_{t-1}	-0.122**	-0.353**	-0.261***	-0.437***	-0.326***	-0.409**
	(0.052)	(0.165)	(0.094)	(0.146)	(0.096)	(0.181)
Estimation Method	OLS	2SLS	OLS	2SLS	OLS	2SLS
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Right Hand Controls	Yes	Yes	Yes	Yes	Yes	Yes
Producer Price	Yes	Yes	No	No	No	No
Not FAS Converted	No	No	Yes	Yes	No	No
Fas Converted	No	No	No	No	Yes	Yes
F-Stat	-	1.1e+04	-	4.5e+04	-	3.5e+04
N	10,285	10,285	10,285	10,285	10,285	10,285
R^2	0.603	0.602	0.604	0.603	0.604	0.603

Note: Robust standard errors clustered at the municipal level are shown in parentheses. The response variable is the log+1 value of homicide. This table shows results for when I average the soil and climate values to calculate the avocado suitability index. Right-hand variables not shown include log population, log male population, percent of the population with primary education completion, logged population 15 years or older and economically active, the cross-sectional 1980-2010 municipal temperature and precipitation, and a dummy accounting for what years El Niño takes place. Columns 1-2 display the OLS and 2SLS results for lagged log real producer prices, respectively. Columns 3-4 show the OLS and 2SLS results for lagged log real non-FAS converted import prices, respectively. Columns 5-6 display the OLS and 2SLS results for lagged log real FAS converted import prices, respectively. For the 2SLS regressions in columns 2, 4, and 6, the interaction term of the avocado suitability index and logged lag real import price is instrumented with the interaction of the avocado suitability index and two-year lagged (in relation to price) weather conditions in the U.S. (average temperature, precipitation, and minimum January temperature). *** is significant at the 1 percent level, ** is significant at the 5 percent level, and * is significant at the 10 percent level.

Table 7: 2SLS Results for Different IVs

	(1) Homicide	(2) Homicide	(3) Homicide	(4) Homicide	(5) Homicide
Index · Ln(ProducerPrice)$_{t-1}$	-0.165***	-0.038*	-0.065***	-0.090***	-0.060**
	(0.062)	(0.022)	(0.025)	(0.034)	(0.024)
Index · USTemp^2	Yes	Yes	No	No	Yes
Index · USPrec^2	Yes	Yes	No	No	Yes
Index · USMin^2	Yes	Yes	No	No	Yes
Index · Log(ProducerPrice)$_{t-2}$	No	No	Yes	Yes	Yes
Index · USTemp	No	Yes	No	Yes	Yes
Index · USPrec	No	Yes	No	Yes	Yes
Index · USMin	No	Yes	No	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes
Right Hand Controls	Yes	Yes	Yes	Yes	Yes
F-Stat	5.1e+03	1.2e+05	2.5e+04	5.4e+04	2.0e+05
N	10,285	10,285	10,285	10,385	10,285
R^2	0.600	0.603	0.603	0.603	0.603

Note: Robust standard errors clustered at the municipal level are shown in parentheses. This table shows the 2SLS results for multiple instrument applications. The response variable is the log+1 value of homicide. Right-hand variables included in the regressions but not shown include log population, log male population, percent of the population with primary education completion, logged population 15 years or older and economically active, the cross-sectional 1980-2010 municipal temperature and precipitation, and a dummy accounting for what years El Niño takes place. Column 1 shows the results of using the interaction of the index with average squared U.S. temperature, precipitation, and minimum January temperature. Column 2 displays the results when using instruments such as the U.S. average temperature, precipitation, and minimum January temperature deviations with the squared U.S. weather deviations. Column 3 shows the 2SLS results when using the interaction of the index with two-year lagged producer price as an instrument. Column 4 shows the results of combining the instrument in column 3 with the three instruments of interaction terms of the index with U.S. temperature, precipitation, and minimum January temperature deviations. Column 5 shows the results of including all seven instruments in the analysis. *** is significant at the 1 percent level, ** is significant at the 5 percent level, and * is significant at the 10 percent level.

Table 8: Main 2SLS Results for Different Weather Time Period IVs

	(1) Homicide	(2) Homicide	(3) Homicide	(4) Homicide	(5) Homicide	(6) Homicide
Index · Ln(Price)_{t-1}	0.165	-0.032	-0.136***	-0.171***	-0.168***	-0.106**
	(0.058)	(0.041)	(0.037)	(0.058)	(0.055)	(0.047)
ProducerPrice	Yes	Yes	Yes	No	No	No
ImportPrice	No	No	No	Yes	Yes	Yes
Weather IV Year	t-1	t-2	t-4	t-1	t-2	t-4
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Right Hand Controls	Yes	Yes	Yes	Yes	Yes	Yes
F-Stat	3.8e+04	5.5e+03	1.0e+04	4.3e+04	6.5e+04	3.3e+04
N	10,285	10,285	10,285	10,385	10,285	10,285
R^2	0.601	0.603	0.601	0.630	0.604	0.604

Note: Robust standard errors clustered at the municipal level are shown in parentheses. The response variable is the log+1 value of homicide. Variables not shown and included in regressions with right-hand controls include log population, log male population, percent of the population with primary education completion, logged population that is 15 years or older and economically active, the cross-sectional 1980-2010 municipal temperature and precipitation, and a dummy accounting for what years El Niño takes place. Columns 1-3 show the results for lagged producer price. Columns 4-6 display the results for lagged import price. For the 2SLS regressions, the interaction term of the avocado suitability index and logged lag real import price is instrumented with the interaction of the avocado suitability index and two-year lagged (in relation to price) weather conditions in the U.S. (average temperature, precipitation, and minimum January temperature). *** is significant at the 1 percent level, ** is significant at the 5 percent level, and * is significant at the 10 percent level.

Table 9: Heterogeneity - Homicide from 2006 to 2019

	(1) Homicide	(2) Homicide	(3) Homicide	(4) Homicide	(5) Homicide	(6) Homicide	(7) Homicide	(8) Homicide
Index · Ln(ProducerPrice)	0.077^{*}	0.076^{*}	0.087^{*}	0.089^{**}	0.442^{***}	0.439^{***}	0.741^{***}	0.745^{***}
	(0.045)	(0.045)	(0.045)	(0.045)	(0.137)	(0.137)	(0.190)	(0.190)
Estimation Method	OLS	OLS	OLS	OLS	2SLS	2SLS	2SLS	2SLS
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Right Hand Controls	Yes	No	Yes	No	Yes	No	Yes	No
F-Stat	-	-	-	-	4.8e+05	5e+05	8.4e+05	8.6e+05
Price Year	t	t	t-1	t-1	t	t	t-1	t-1
N	6,192	6,192	6,192	6,192	6,192	6,192	6,192	6,192
R^2	0.589	0.586	0.589	0.586	0.585	0.581	0.571	0.568

Note: Robust standard errors clustered at the municipal level are shown in parentheses. The response variable is the log+1 value of homicide. This table applies the same methods as Table 2; however, this table does not include data before 2006. Variables not shown and included in regressions with right-hand controls include log population, log male population, percent of the population with primary education completion, logged population that is 15 years or older and economically active, the cross-sectional 1980-2010 municipal temperature and precipitation, and a dummy accounting for what years El Niño takes place. Columns 1-2 and 5-6 evaluate producer price at time t, while columns 3-4 and 7-8 measure producer price at time t-1. The F-statistic in each 2SLS regression refers to the Angrist-Pischke F-statistic of excluded instruments. In columns 5-8, the interaction term of the avocado suitability index and logged real producer price is instrumented with the interaction of the avocado suitability index and two-year lagged (in relation to price) weather conditions in the U.S. (average temperature, precipitation, and minimum January temperature). *** is significant at the 1 percent level, ** is significant at the 5 percent level, and * is significant at the 10 percent level.

Table 10: Results for Crime from 2011-2017

	(1) Property	(2) Property	(3) Cattle Theft	(4) Cattle Theft	(5) Threats	(6) Threats	(7) Gun	(8) Gun
Index · $\text{Ln}(\text{ProducerPrice})_{t-1}$	0.162***	0.208*	0.157**	0.217*	0.181***	0.310**	0.184**	0.359***
	(0.063)	(0.114)	(0.065)	(0.124)	(0.069)	(0.130)	(0.084)	(0.139)
Estimation Method	OLS	2SLS	OLS	2SLS	OLS	2SLS	OLS	2SLS
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Right Hand Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F-Stat	-	4e+03	-	4e+03	-	4e+03	-	4e+03
N	2,316	2,316	2,316	2,316	2,316	2,316	2,316	2,316
R^2	0.885	0.885	0.719	0.719	0.832	0.831	0.673	0.671

Note: Robust standard errors clustered at the municipal level are shown in parentheses. All response variables are the log+1 values. Right-hand control variables not shown in the table include the El Niño dummy and municipality population from the 2010 INEGI census. Columns 1-2 show the OLS and 2SLS results for property crime, respectively. Columns 3-4 show the OLS and 2SLS results for cattle theft, respectively. Columns 5-6 show the OLS and 2SLS results for reported threats, respectively. Columns 7-8 show the OLS and 2SLS results for homicide by gunfire, respectively. The F-statistic in each 2SLS regression refers to the Angrist-Pischke F-statistic of excluded instruments. In columns 5-8, the interaction term of the avocado suitability index and logged real producer price is instrumented with the interaction of the avocado suitability index and two-year lagged (in relation to price) weather conditions in the U.S. (average temperature, precipitation, and minimum January temperature). *** is significant at the 1 percent level, ** is significant at the 5 percent level, and * is significant at the 10 percent level.

Table 11: Additional Results for Crime from 2011-2017

	(1) Rape	(2) Rape	(3) Highway	(4) Highway	(5) Kidnapping	(6) Kidnapping
Index · $\text{Ln}(\text{ProducerPrice})_{t-1}$	0.018	0.053	-0.003	0.020	0.016	0.012
	(0.057)	(0.099)	(0.075)	(0.057)	(0.035)	(0.073)
Estimation Method	OLS	2SLS	OLS	2SLS	OLS	2SLS
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Right Hand Controls	Yes	Yes	Yes	Yes	Yes	Yes
F-Stat	-	4e+03	-	4e+03	-	4e+03
N	2,316	2,316	2,316	2,316	2,316	2,316
R^2	0.682	0.682	0.538	0.538	0.458	0.458

Note: Robust standard errors clustered at the municipal level are shown in parentheses. All response variables are in log+1 values. Right-hand control variables not shown in the table include the El Niño dummy and municipality population from the 2010 INEGI census. Columns 1-2 show the OLS and 2SLS results for rape, respectively. Columns 3-4 show the OLS and 2SLS results for highway robberies, respectively. Columns 5-6 show the OLS and 2SLS results for kidnappings, respectively. The F-statistic in each 2SLS regression refers to the Angrist-Pischke F-statistic of excluded instruments. In columns 5-8, the interaction term of the avocado suitability index and logged real producer price is instrumented with the interaction of the avocado suitability index and two-year lagged (in relation to price) weather conditions in the U.S. (average temperature, precipitation, and minimum January temperature). *** is significant at the 1 percent level, ** is significant at the 5 percent level, and * is significant at the 10 percent level.

8 Appendix

Table A1: Summary Statistics for 1990 and 2000 Differences

Variable	Observations	Mean	Std. Dev.	Min	Max
Secondary Jobs	449	-95.644	628.097	-3,247	2,884
Index	449	2.332	1.933	0	6
Population	449	3,997	6,597.08	-12,917	45,954
Male Population	449	1,840.817	3,221.072	-6,330	22,738
Economically Active Persons	449	2,358.927	2,969.826	-1,167	19,410
Primary Education Completion	449	988.187	1,006.916	-989	7,192
Homes with Flooring	449	1,479.98	1,583.766	-217	11,214
Homes with Running Water	449	1,398.107	1,526.195	-600	11,243
Homes with Electricity	449	1,660.252	1,674.899	-123	11,353
Temperature	449	19.566	4.052	10.9	29.5
Precipitation	449	868.863	271.447	302.8	2,267.9

Note: See data section for definitions of variables.

Table A2: Summary Statistics 2 for 2011 - 2017 Analysis

Variable	Observations	Mean	Std. Dev.	Min	Max
Panel-level municipal variables					
Property	2,316	26.352	36.757	0	250
Cattle Theft	2,316	2.595	4.555	0	48
Reported Threats	2,316	5.562	12.736	0	175
Rape	2,316	1.935	2.898	0	26
Highway Robbery	2,316	0.703	2.879	0	51
Kidnapping	2,316	0.300	0.853	0	14
El Niño	2,316	0.375	0.484	0	1
Cross-sectional municipal variables					
Index 1	428	2.186	1.961	0	6
Mexico 1980 - 2010 Temperature Avg. (Celcius)	438	20.248	3.953	10.9	29.5
Mexico 1980 - 2010 Precipitation Avg. (Celcius)	438	884.335	282.47	302.8	2,267.9
Population 2010 Census	438	27,324.77	21,689.73	2,082	99,576
Annual-level variables					
Real Mexico Producer Price (USD/Ton)	7	640.874	71.054	561.114	805.639
Lagged Avg. Temp. Deviation (Degrees)	7	-0.428	0.742	-1.2	0.73
Lagged Avg. Prec. Deviation (Inches)	7	-0.022	1.050	-1.33	2.31
Lagged Avg. Minimum Temp Deviation(Degrees)	7	2.297	1.880	-0.686	5

Note: See data section for definitions of variables.

Table A3: Soil Characteristics

Variable	Classification	Value	Value Description
Andosol	Highly Suitable	3	7.7 percent organic matter, formed in volcanic tephra
Luvisol	Suitable	2	Favorable textural and structural characteristics for good water physical properties
Phaeozems	Suitable	2	Favorable textural and structural characteristics for good water physical properties
Kastanozem	Suitable	2	High Base Saturation and granular structure
Gleysol	Suitable	2	High base saturation, good depth, crumb shape, low hydraulic conductivity
Solonchak	Suitable	2	High base saturation, deep depth, crum to blocky shape, low hydraulic conductivity
Calcisol	Suitable	2	High base saturation, depth within 100 cm, blocky
Chernozem	Suitable	2	High base saturation, deep depth, blocky
Cambisol	Low Suitability	1	2.3 percent organic matter
Regosol	Low Suitability	1	Low cation exchange capacity
Arenosol	Low Suitability	1	Low base saturation, varying depth, $<$ 15 percent clay content
Histosol	Low Suitability	1	Varying base saturation, varying aggregate shape, varying depth, very low hydraulic conductivity
Fluvisol	Low Suitability	1	Varying vase saturation, depth $>$ 50 cm, low hydraulic conductivity
Vertisol	No Suitability	0	Clayey texture and Massive Structure
Leptosol	No Suitability	0	Shallow depth profile ($<$ 25 cm)
Planosol	No Suitability	0	Varying base saturation, varying depth, massive shape, low hydraulic conductivity, high clay texture
Durisol	No Suitability	0	Base saturation $>$ 50 percent, within 100 cm depth, typically massive shape

Note: Soil information comes from Dubrovina and Bautista (2014) and ISRIC World Soil Information. This list includes soil that is the most common among at least one municipality in the analysis.

Table A4: First Stage Results for Table 2

	(1) Index · $\text{Ln(ProducerPrice)}_t$	(2) Index · $\text{Ln(ProducerPrice)}_{t-1}$
Index · USTemp	0.009***	0.010***
	(0.000)	(0.000)
Index · USPrec	-0.051***	-0.055***
	(0.000)	(0.000)
Index · USMin	0.014***	0.023***
	(0.000)	(0.000)
F-Stat	2.3e+04	1.4e+04
Year FE	Yes	Yes
Municipality FE	Yes	Yes
Right Hand Controls	Yes	Yes
N	10,285	10,285

Note: Robust standard errors clustered at the municipal level are shown in parentheses. This table displays the first-stage results for Table 2. Variables not shown and included in regressions with right-hand controls include log population, log male population, percent of the population with primary education completion, logged population that is 15 years or older and economically active, the cross-sectional 1980-2010 municipal temperature and precipitation, and a dummy accounting for what years El Niño takes place. Columns 1-3 show the results for lagged producer price. Column 1 represents the first-stage result for price in year t, while column 2 shows the first-stage result or price in year t-1. *** is significant at the 1 percent level, ** is significant at the 5 percent level, and * is significant at the 10 percent level.

Table A5: First Stage Results of Table 7

	(1)
	Index $\cdot$ Ln(ProducerPrice)$_{t-1}$
Index $\cdot$ USTemp	0.084***
	(0.000)
Index $\cdot$ USPrec	-0.093***
	(0.000)
Index $\cdot$ USMin	0.021***
	(0.000)
Index $\cdot$ USTemp^2	-0.35***
	(0.000)
Index $\cdot$ USPrec^2	-0.043***
	(0.000)
Index $\cdot$ USMin^2	0.004***
	(0.000)
Index $\cdot$ Ln(ProducerPrice)$_{t-2}$	0.522***
	(0.001)
F-Stat	2.0e+05
Year FE	Yes
Municipality FE	Yes
Right Hand Controls	Yes
N	10,285

Note: Robust standard errors clustered at the municipal level are shown in parentheses. This table displays the first-stage results for Table 7. Variables not shown include right-hand controls such as log population, log male population, percent of the population with primary education completion, logged population that is 15 years or older and economically active, the cross-sectional 1980-2010 municipal temperature and precipitation, and a dummy accounting for what years El Niño takes place. Columns 1-3 show the results for lagged producer price. Column 1 specifically displays the first-stage result of column 5 of Table 7. *** is significant at the 1 percent level, ** is significant at the 5 percent level, and * is significant at the 10 percent level.

Table A6: Table 2 2SLS Results with Different IV Combinations

	(1) Homicide	(2) Homicide	(3) Homicide	(4) Homicide	(5) Homicide
Index · Ln(ProducerPrice)$_{t-1}$	-0.527***	-0.143*	-0.200***	-0.312***	-0.124**
	(0.111)	(0.074)	(0.073)	(0.076)	(0.055)
Index · USTemp	Yes	No	Yes	Yes	No
Index · USPrec	No	Yes	Yes	No	Yes
Index · USMin	No	No	No	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes
Right Hand Controls	Yes	Yes	Yes	Yes	Yes
F-Stat	5.3e+03	6.7e+03	4.1e+03	1.1e+04	1.4e+04
N	10,285	10,285	10,285	10,385	10,285
R^2	0.549	0.632	0.598	0.643	0.602

Note: Robust standard errors clustered at the municipal level are shown in parentheses. The response variable is the log+1 value of homicide. Variables not shown include right-hand controls such as log population, log male population, percent of the population with primary education completion, logged population that is 15 years or older and economically active, the cross-sectional 1980-2010 municipal temperature and precipitation, and a dummy accounting for what years El Niño takes place. Column displays results for when only using the interaction of Index with average U.S. temperature deviation as an instrument. Column 2 shows the results of only using the ineraction of Index with the average U.S. precipitation as an instrument. Column 3 displays the results of using the 2 instruments from columns 1 and 2. Column 4 shows the result of instrumenting the 2 interaction terms of Index with the average U.S. temperature deviations and average minimum January deviation. Column 5 shows the result of including the 2 interaction instruments of Index with U.S. average precipitation deviations and U.S. average January minimum temperature deviations. *** is significant at the 1 percent level, ** is significant at the 5 percent level, and * is significant at the 10 percent level.

Table A7: 2SLS Results of Table 2 without Clustering Robust Standard Errors

	(1) Homicide	(2) Homicide	(3) Homicide	(4) Homicide
Index · Ln(ProducerPrice)	-0.185***	-0.197***	-0.162***	-0.160***
	(0.054)	(0.054)	(0.046)	(0.046)
Year FE	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes
Right Hand Controls	Yes	No	Yes	No
F-Stat	316.72	317.64	279.06	277.44
Price Year	t	t	t-1	t-1
N	10,285	10,285	10,285	10,285
R^2	0.600	0.636	0.633	0.636

Note: This table shows the 2SLS results of Table 2 when standard errors are not clustered at the municipal level. The response variable is the log+1 value of homicide. Variables not shown among regressions including right-hand controls are log population, log male population, percent of the population with primary education completion, logged population that is 15 years or older and economically active, the cross-sectional 1980-2010 municipal temperature and precipitation, and a dummy accounting for what years El Niño takes place. *** is significant at the 1 percent level, ** is significant at the 5 percent level, and * is significant at the 10 percent level.

Table A8: Table 2 Results with Inverse Hyperbolic Sine of Homicide

	(1) Homicide	(2) Homicide	(3) Homicide	(4) Homicide	(5) Homicide	(6) Homicide	(7) Homicide	(8) Homicide
Index · Ln(ProducerPrice)	-0.054**	-0.049*	-0.054**	-0.039*	-0.208**	-0.222***	-0.186***	-0.184***
	(0.027)	(0.027)	(0.022)	(0.022)	(0.082)	(0.083)	(0.069)	(0.057)
Estimation Method	OLS	OLS	OLS	OLS	2SLS	2SLS	2SLS	2SLS
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Right Hand Controls	Yes	No	Yes	No	Yes	No	Yes	No
F-Stat	-	-	-	-	2.3e+04	5.3e+04	1.4e+04	5.4e+04
Price Year	t	t	t-1	t-1	t	t	t-1	t-1
N	10,285	10,285	10,285	10,285	10,285	10,285	10,285	10,285
R^2	0.602	0.600	0.603	0.600	0.600	0.596	0.600	0.596

Note: Robust standard errors clustered at the municipal level are shown in parentheses. This table displays the results of Table 2, but the response variable is the inverse hyperbolic sine of homicide instead of the log+1 value. Variables not shown and included in regressions with right-hand controls include log population, log male population, percent of the population with primary education completion, logged population that is 15 years or older and economically active, the cross-sectional 1980-2010 municipal temperature and precipitation, and a dummy accounting for what years El Niño takes place. Columns 1-2 and 5-6 evaluate producer price at time t, while columns 3-4 and 7-8 measure producer price at time t-1. The F-statistic in each 2SLS regression refers to the Angrist-Pischke F-statistic of excluded instruments. In columns 5-8, the interaction term of the avocado suitability index and logged real producer price is instrumented with the interaction of the avocado suitability index and two-year lagged (in relation to price) weather conditions in the U.S. (average temperature, precipitation, and minimum January temperature). *** is significant at the 1 percent level, ** is significant at the 5 percent level, and * is significant at the 10 percent level.

Table A9: Results with Third Index Method

	(1) Homicide	(2) Homicide	(3) Homicide	(4) Homicide	(5) Homicide	(6) Homicide
Index · Ln(Price)_{t-1}	-0.061**	-0.177**	-0.131***	-0.219***	-0.163***	-0.204**
	(0.026)	(0.083)	(0.047)	(0.073)	(0.048)	(0.090)
Producer Price	Yes	Yes	No	No	No	No
Not FAS Converted	No	No	Yes	Yes	No	No
FAS Converted	No	No	No	No	Yes	Yes
Estimation Method	OLS	2SLS	OLS	2SLS	OLS	2SLS
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Right Hand Controls	Yes	Yes	Yes	Yes	Yes	Yes
F-Stat	-	1.1e+04	-	4.5e+04	-	3.5e+04
N	10,285	10,285	10,285	10,285	10,285	10,285
R^2	0.646	0.602	0.604	0.603	0.604	0.603

Note: Robust standard errors clustered at the municipal level are shown in parentheses. This table shows results for when I sum the soil and climate values to calculate the avocado suitability index. Right-hand variables not shown include log population, log male population, percent of the population with primary education completion, logged population 15 years or older and economically active, the cross-sectional 1980-2010 municipal temperature and precipitation, and a dummy accounting for what years El Niño takes place. Columns 1-2 display the OLS and 2SLS results for lagged log real producer prices, respectively. Columns 3-4 show the OLS and 2SLS results for lagged log real non-FAS converted import prices, respectively. Columns 5-6 display the OLS and 2SLS results for lagged log real FAS converted import prices, respectively. For the 2SLS regressions in columns 2, 4, and 6, the interaction term of the avocado suitability index and logged lag real import price is instrumented with the interaction of the avocado suitability index and two-year lagged (in relation to price) weather conditions in the U.S. (average temperature, precipitation, and minimum January temperature). *** is significant at the 1 percent level, ** is significant at the 5 percent level, and * is significant at the 10 percent level.

Table A10: Results of Table 10 with Inverse-Hyperbolic Sine Left-Hand Values

	(1) Property	(2) Property	(3) Cattle Theft	(4) Cattle Theft	(5) Threats	(6) Threats	(7) Gun	(8) Gun
Index $\cdot$ Ln(ProducerPrice)$_{t-1}$	0.190***	0.244*	0.199**	0.273*	0.189**	0.178*	0.218**	0.427***
	(0.073)	(0.134)	(0.083)	(0.157)	(0.083)	(0.100)	(0.105)	(0.174)
Estimation Method	OLS	2SLS	OLS	2SLS	OLS	2SLS	OLS	2SLS
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Right Hand Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F-Stat	-	4e+03	-	4e+03	-	3.1e+06	-	4e+03
N	2,316	2,316	2,316	2,316	2,316	2,316	2,316	2,316
R^2	0.870	0.870	0.712	0.711	0.829	0.829	0.662	0.661

Note: Robust standard errors clustered at the municipal level are shown in parentheses. This table uses the same procedure as Table 10; however, the table displays the results of the inverse hyperbolic sine values of the response variable instead of the log+1 values. Right-hand control variables not shown in the table include the El Niño dummy and municipality population from the 2010 INEGI census. Columns 1-2 show the OLS and 2SLS results for property crime, respectively. Columns 3-4 show the OLS and 2SLS results for cattle theft, respectively. Columns 5-6 show the OLS and 2SLS results for reported threats, respectively. Columns 7-8 show the OLS and 2SLS results for homicide by gunfire, respectively. The F-statistic in each 2SLS regression refers to the Angrist-Pischke F-statistic of excluded instruments. In columns 5-8, the interaction term of the avocado suitability index and logged real producer price is instrumented with the interaction of the avocado suitability index and two-year lagged (in relation to price) weather conditions in the U.S. (average temperature, precipitation, and minimum January temperature). *** is significant at the 1 percent level, ** is significant at the 5 percent level, and * is significant at the 10 percent level.

Table A11: 2SLS Results of Table 7 Using Alternative IVs

	(1) Property	(2) Property	(3) Cattle Theft	(4) Cattle Theft	(5) Threats	(6) Threats	(7) Gun	(8) Gun
Index · Ln(ProducerPrice)$_{t-1}$	0.323***	0.186***	0.244**	0.182***	0.436***	0.211***	0.490***	0.251**
	(0.108)	(0.063)	(0.110)	(0.062)	(0.133)	(0.069)	(0.174)	(0.101)
Index · Ln(ProducerPrice)$_{t-2}$	Yes	No	Yes	No	Yes	No	Yes	No
Index · USTemp^2	No	Yes	No	Yes	No	Yes	No	Yes
Index · USPrec^2	No	Yes	No	Yes	No	Yes	No	Yes
Index · USMin^2	No	Yes	No	Yes	No	Yes	No	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Right Hand Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F-Stat	7.3e+04	2.2e+07	7.3e+04	2.2e+07	7.3e+04	2.2e+07	7.3e+07	2.2e+07
N	2,316	2,316	2,316	2,316	2,316	2,316	2,316	2,316
R^2	0.884	0.885	0.719	0.719	0.830	0.832	0.660	0.662

Note: Robust standard errors clustered at the municipal level are shown in parentheses. This table shows the results of using alternative instrument methods. Each left-hand variable is in log+1 value. Right-hand control variables not shown in the table include the El Niño dummy and municipality population from the 2010 INEGI census. These methods include using the interaction instrument of the index and t-2 producer price or the squared U.S. average temperature, precipitation, and minimum January deviations. Columns 1-2 show the 2SLS results for property crime, columns 3-4 show the 2SLS results for cattle theft, columns 5-6 show the 2SLS results for reported threats, and columns 7-8 show 2SLS results for homicide by gunfire. The F-statistic in each 2SLS regression refers to the Angrist-Pischke F-statistic of excluded instruments. In columns 5-8, the interaction term of the avocado suitability index and logged real producer price is instrumented with the interaction of the avocado suitability index and two-year lagged (in relation to price) weather conditions in the U.S. (average temperature, precipitation, and minimum January temperature). *** is significant at the 1 percent level, ** is significant at the 5 percent level, and * is significant at the 10 percent level.

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